

Statistical Inference (MCMC)

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Announcements

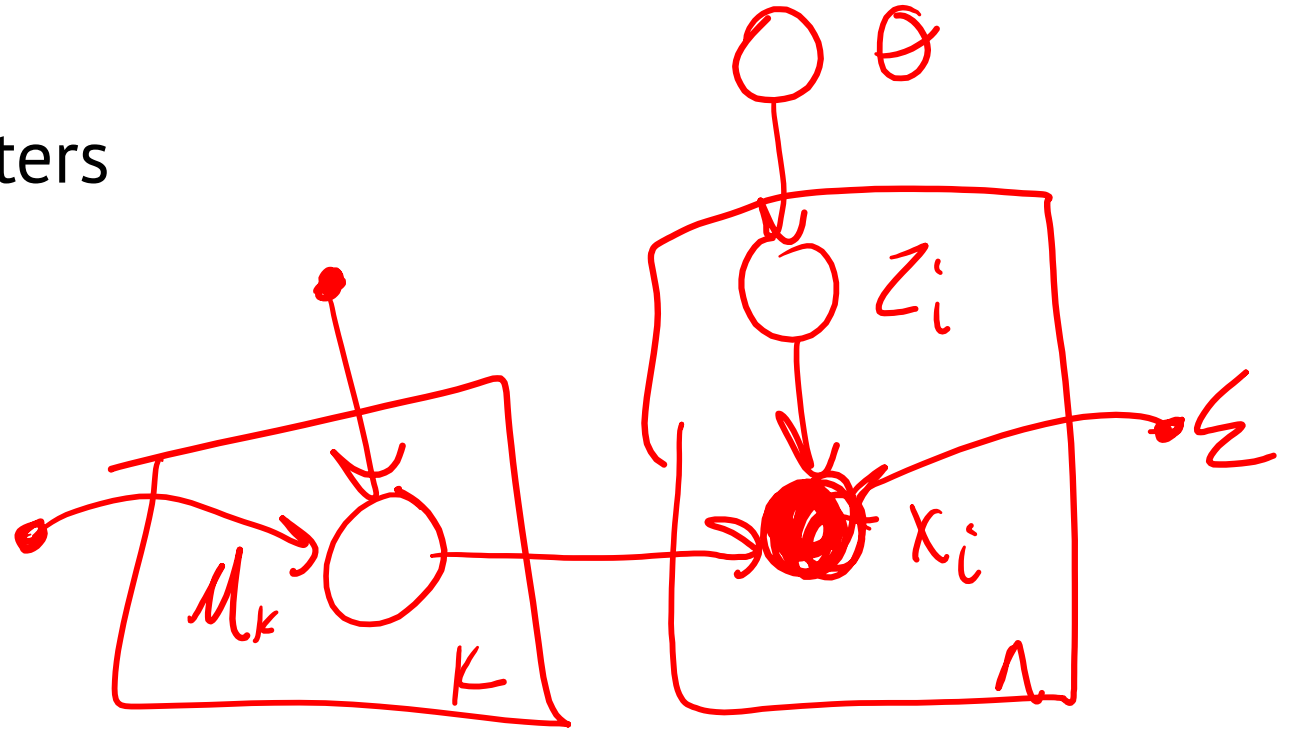
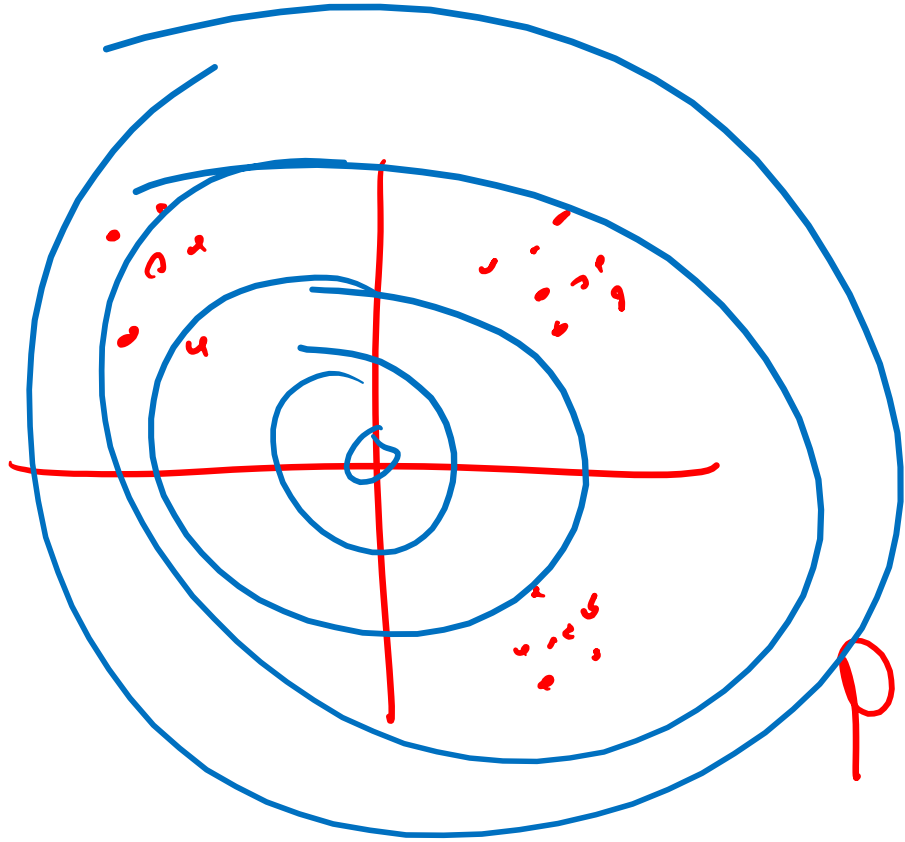
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Inference

- There are several quantities we might want to infer given a probabilistic model
 - Maximum likelihood estimates: $\tilde{\pi}_{MLE} = \arg \max_{\pi} p(X|\pi)$
 - Posterior modes: $\tilde{\pi}_{MAP} = \arg \max_{\pi} p(\pi|X) = \arg \max_{\pi} p(X|\pi)p(\pi)$
 - Marginal probabilities: $p(\pi)$
- And, we have covered inference methods to compute these
 - Analytical solution (Beta-Bernoulli), MLE
 - Sum-product/max-product algorithms

Bayesian Mixture Model

- We have n observations
- Each from one of k Gaussian clusters

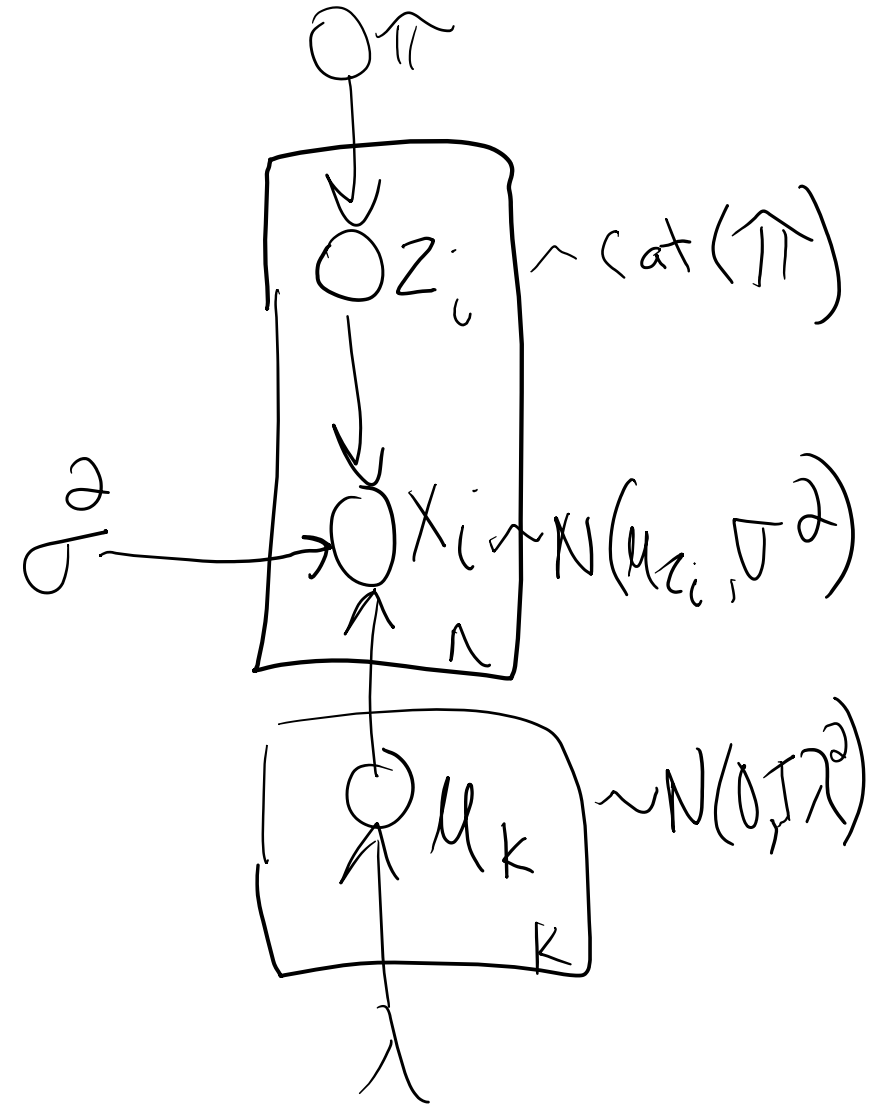
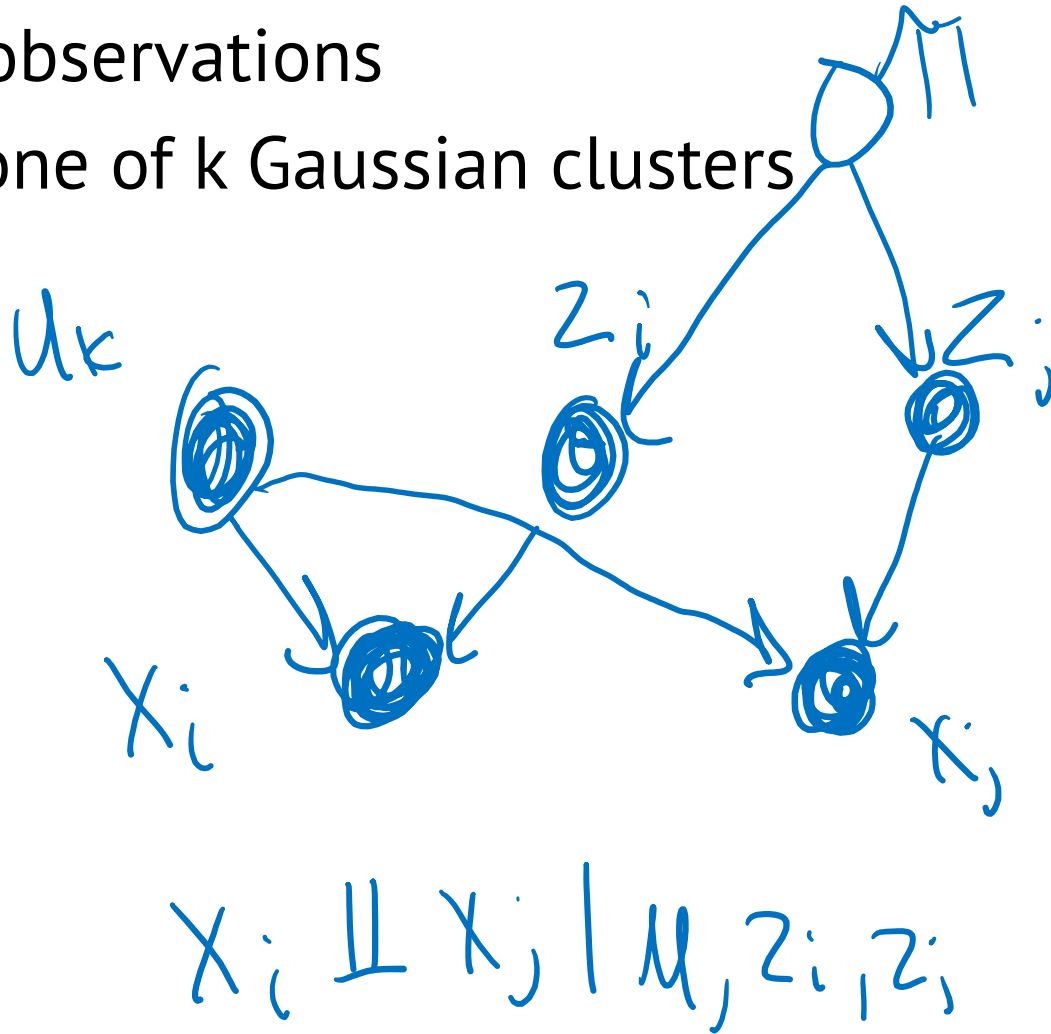


$$p(\mu_k; 0, 10I)$$

$$p(x_i | \mu, z_i) \sim N(\mu_{z_i}, \Sigma)$$

Bayesian Mixture Model

- We have n observations
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Bayesian Mixture Model

- Suppose $k = 3$
- Posterior for mixture components

Bayesian Mixture Model

- Suppose $k = 3$
- Posterior for mixture components
- Numerator?

$$p(\mu_1, \mu_2, \mu_3 | \mathbf{x}) = \frac{p(\mu_1, \mu_2, \mu_3, \mathbf{x})}{\int \int \int p(\mu'_1, \mu'_2, \mu'_3, \mathbf{x}) d\mu'_1 d\mu'_2 d\mu'_3}$$

Bayesian Mixture Model

- Suppose $k = 3$
- Posterior for mixture components
- Numerator

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$$p(\mu_1) p(\mu_2) p(\mu_3) \prod_{i=1}^n \sum_{k=1}^K \pi_k p(x_i | \mu_k)$$

$$p(\mathbf{x}) = \int \int \int p(\mu'_1) p(\mu'_2) p(\mu'_3) \prod_{i=1}^n \sum_{k=1}^K \pi_k p(x_i | \mu'_k) d\mu'_1 d\mu'_2 d\mu'_3$$

Why is inference difficult?

- Norm. constant $p(\mathbf{x}) = \int \int \int p(\mu'_1) p(\mu'_2) p(\mu'_3) \prod_{i=1}^n \sum_{k=1}^K \pi_k p(x_i | \mu'_k) d\mu'_1 d\mu'_2 d\mu'_3$

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- Sum over $\mathbf{z}$ instead of k $p(\mathbf{x}) = \sum_{\mathbf{z}} p(\mathbf{z}) \int \int \int p(\mu'_1) p(\mu'_2) p(\mu'_3) \prod_{i=1}^n p(x_i | \mu'_{z_i}) d\mu'_1 d\mu'_2 d\mu'_3$

Why is inference difficult?

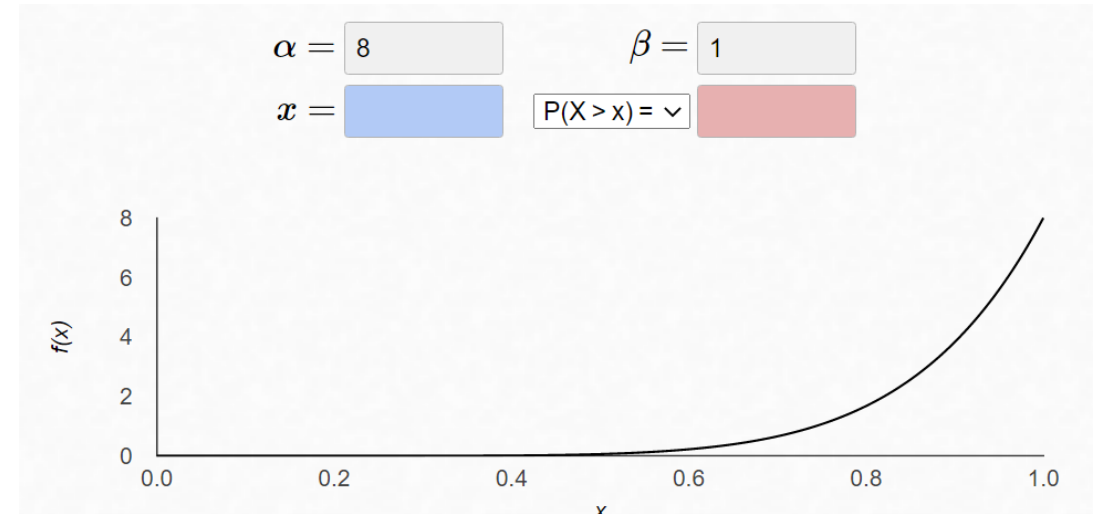
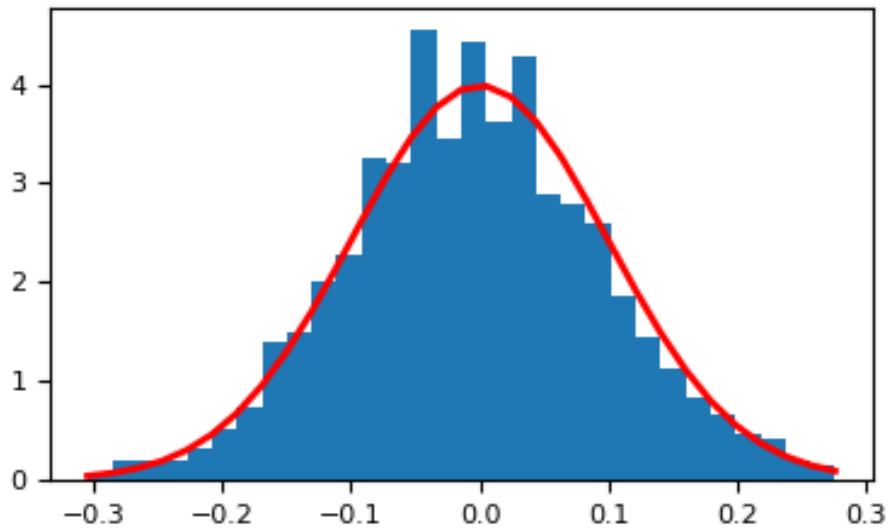
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- Partition data according to $\mathbf{z}$ $p(\mathbf{x}) = \sum_{\mathbf{z}} p(\mathbf{z}) \prod_{k=1}^3 \left(\int p(\mu'_k) \prod_{\{i: z_i=k\}} p(x_i | \mu'_k) d\mu'_k \right)$

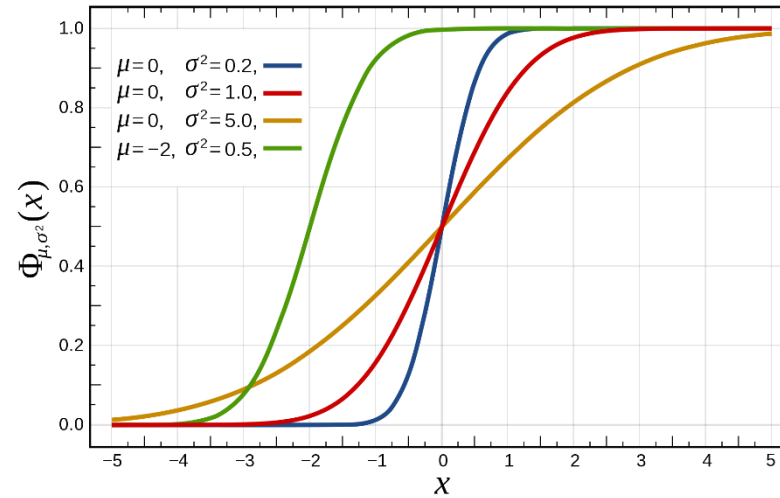
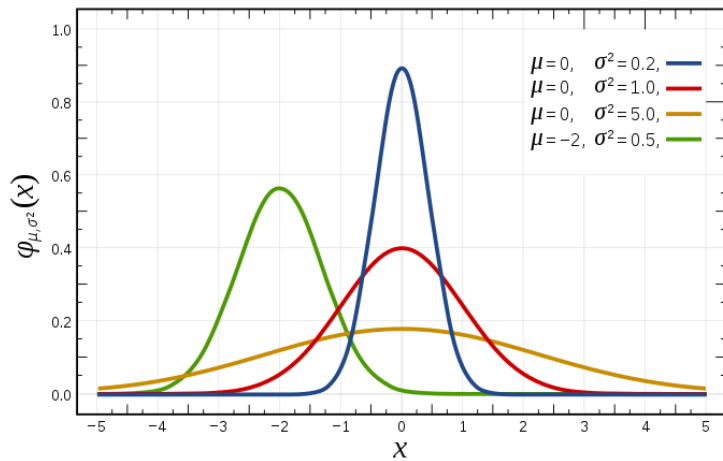
Let's try sampling

- An alternative way to categorize a distribution
 - Approximate distribution with density of samples



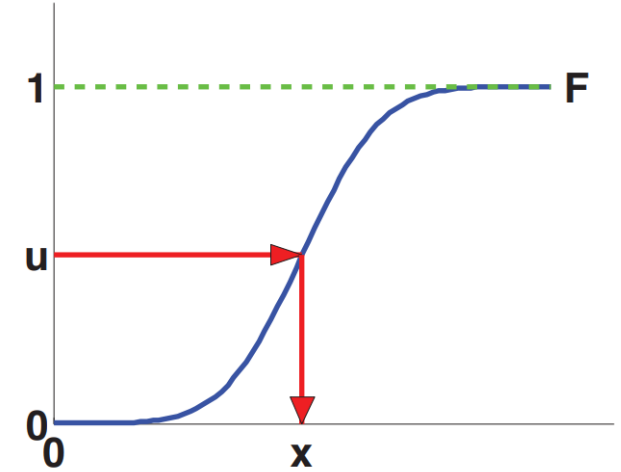
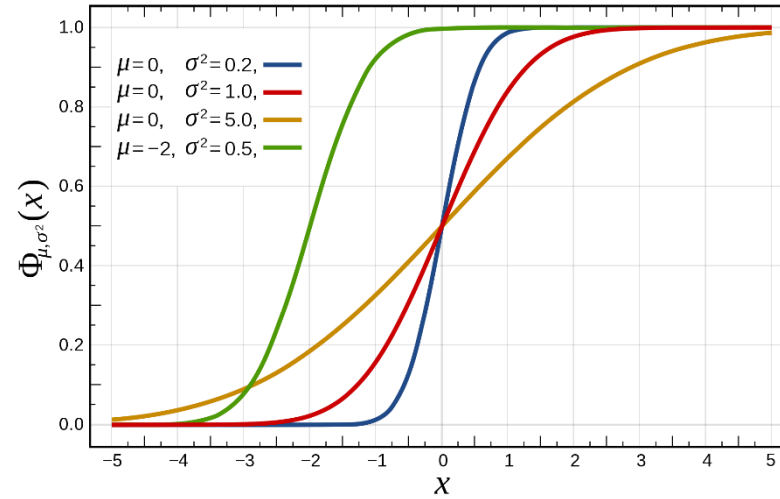
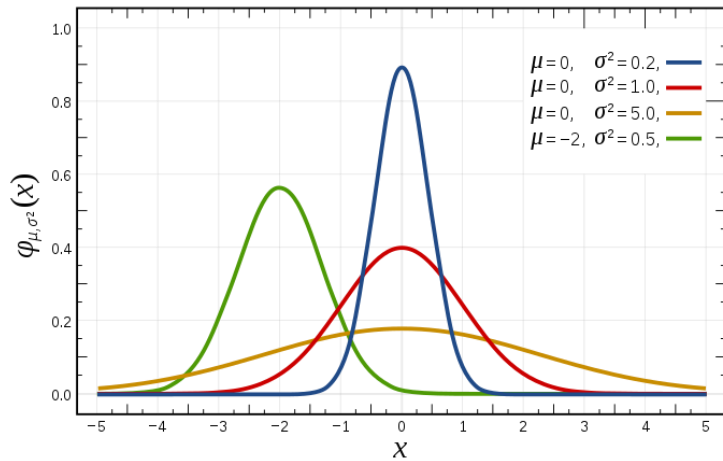
Sampling from density

- How do we sample from a Normal density?



Sampling from density

- How do we sample from a Normal density?



- [demo]

Also:

<https://github.com/numpy/numpy/blob/2afa142ae6ee121f6c75f28403526e35473ee6d5/numpy/random/mtrand/randomkit.c#L619> if interested in code

Sampling

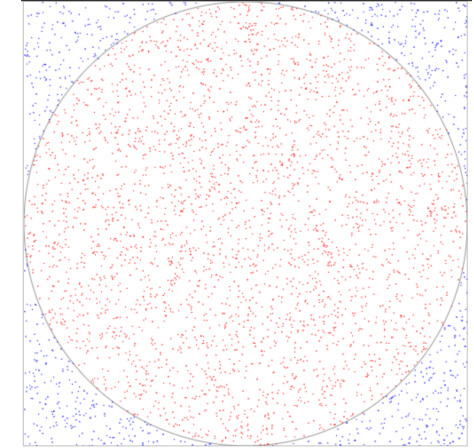
- What if we have a particular complex distribution?
- How about a using *Monte Carlo* approximation?

$$E[f(z)] = \int f(z)p(z) \, dz$$

$$E[p(\theta|X)] = \int_{x_1} \dots \int_{x_n} \theta p(\theta|X) d\theta$$

Sampling approach (Monte Carlo)

- Monte Carlo turns estimation into a task of simulation and probabilistic choices
- Consider estimating π (the number)
- We draw a square and inscribe a circle inside
- The ratio of their areas is $\frac{C}{S} = \frac{\pi\left(\frac{d}{2}\right)^2}{d^2} \Rightarrow \pi = \frac{4C}{S}$
- Now, approximate C and S by randomly dropping rice in the area



Sampling Mixture Models

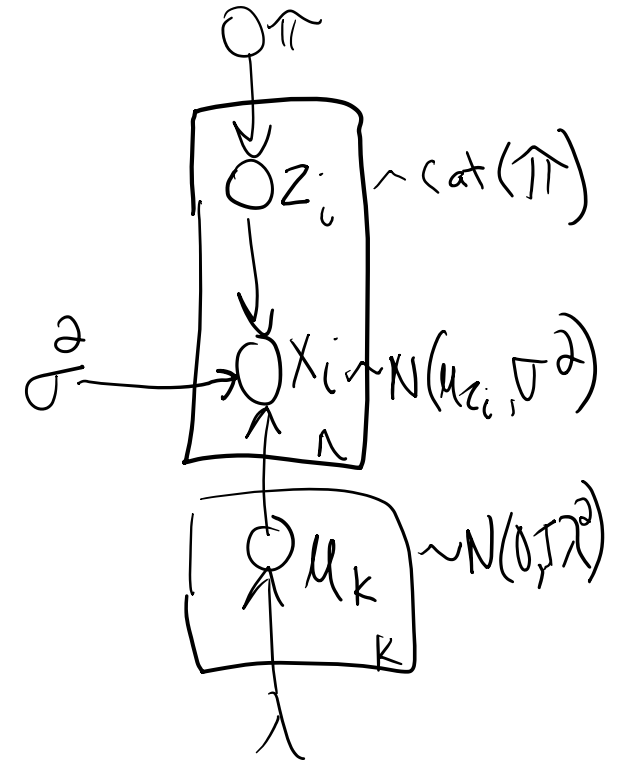
- Numerator is easy to compute

$$p(\mu_1)p(\mu_2)p(\mu_3) \prod_{i=1}^n \sum_{k=1}^K \pi_k p(x_i | \mu_k)$$

- Denominator is difficult

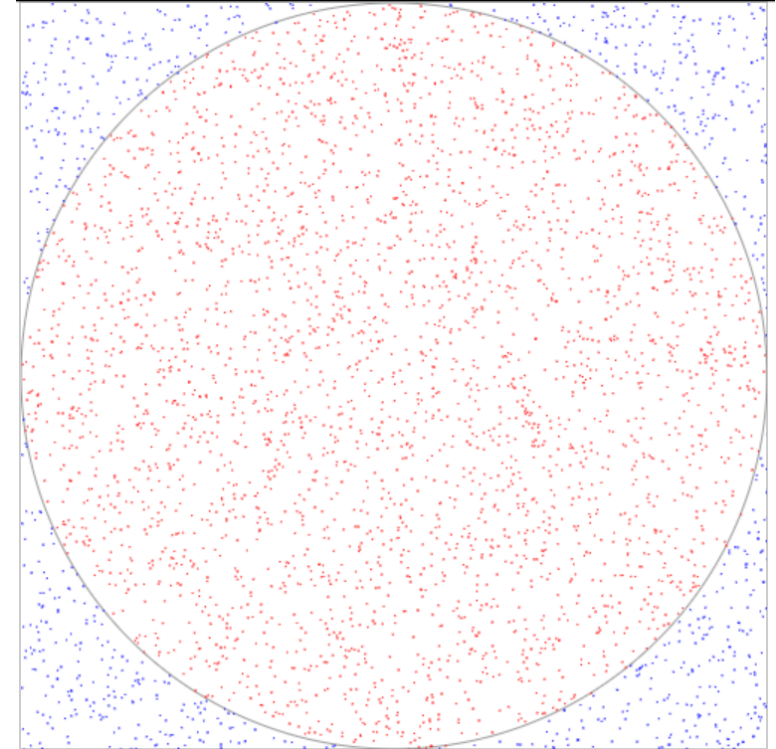
$$p(\mathbf{x}) = \sum_{\mathbf{z}} p(\mathbf{z}) \prod_{k=1}^3 \left(\int p(\mu'_k) \prod_{\{i: z_i=k\}} p(x_i | \mu'_k) d\mu'_k \right)$$

- We have $p(\mu|x) = \frac{1}{Z} p(\mu, x)$ where Z is hard to compute but $p(\mu, x)$ is easy to evaluate



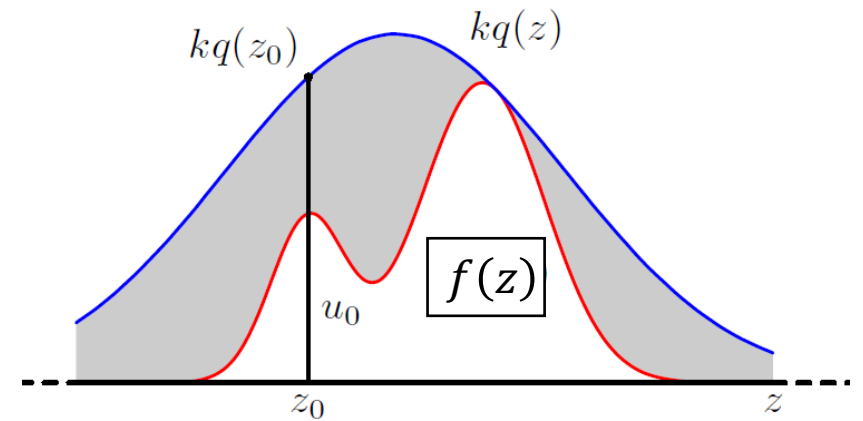
Rejection sampling

- Idea: generate points from a tractable distribution, and reject some points in order to sample from $f(z)$
- Assume:
 - we have access to an unnormalized density $f(z)$
 - We can generate points from a distribution with density proportional to $k * q(z)$ such that $f(z) \leq k * q(z)$, $\forall z$



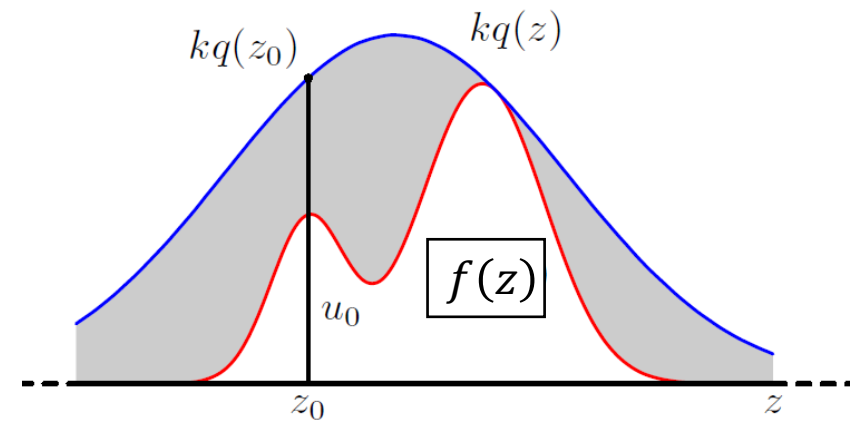
Rejection sampling

- Sample z_0 from the proposal distribution $q(z)$



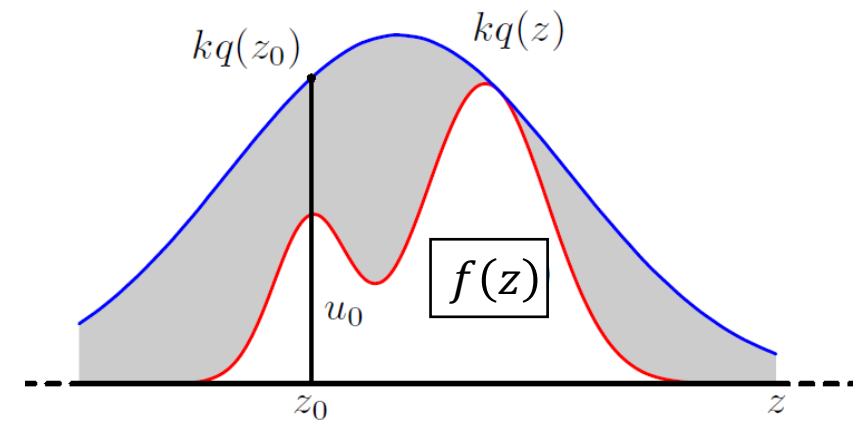
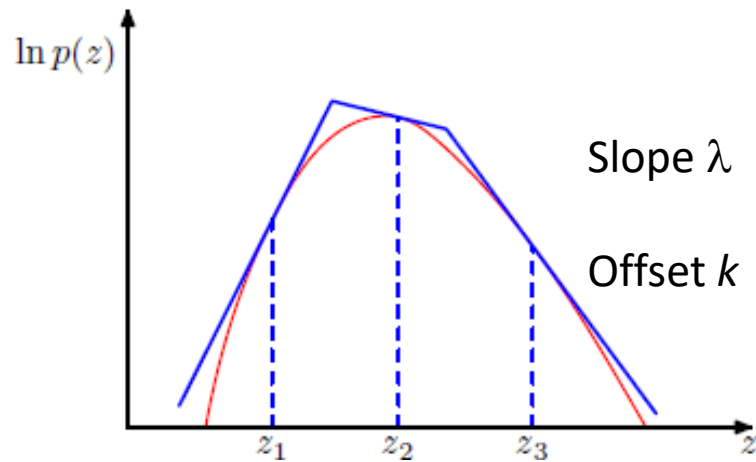
Rejection sampling

- Sample z_0 from the proposal distribution $q(z)$
- Accept z_0 with probability $\frac{f(z)}{k \cdot q(z)}$



Rejection sampling

- Sample z_0 from the proposal distribution $q(z)$
- Accept z_0 with probability $\frac{f(z)}{k * q(z)}$
- But... Must construct $k * q(z)$ carefully
 - Very wasteful for high dimensions



See adaptive rejection sampling

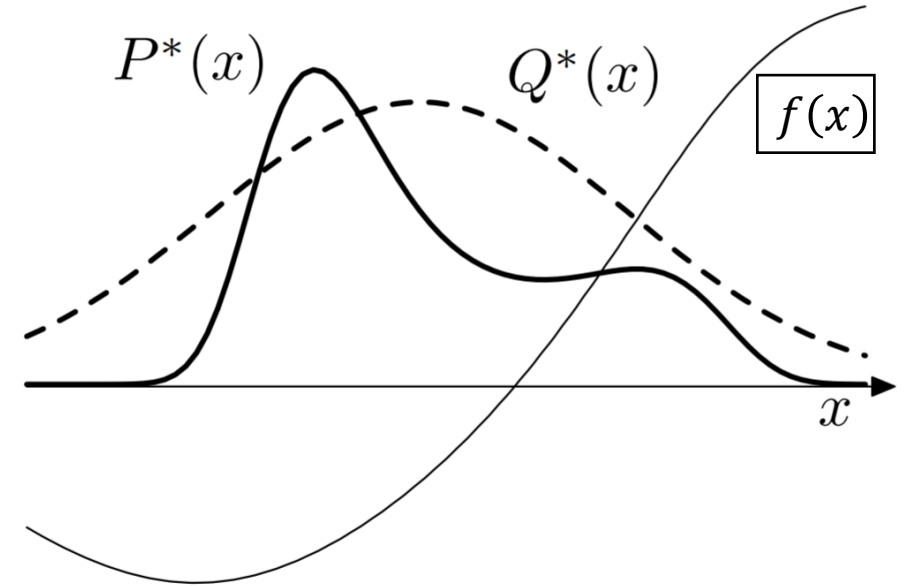
Rejection Sampling

Algorithm 27.2 Rejection sampling to draw L independent samples from $p(x) = p^*(x)/Z$.

```
1: Given  $p^*(x)$  and  $q(x)$ , find  $M$  such that  $p^*(x)/q(x) \leq M$  for all  $x$ .
2: for  $l = 1$  to  $L$  do
3:   repeat
4:     Draw a candidate sample  $x^{cand}$  from  $q(x)$ .
5:     Let  $a = \frac{p^*(x^{cand})}{Mq(x^{cand})}$ 
6:     Draw a value  $u$  uniformly between 0 and 1.
7:   until  $u \leq a$ 
8:    $x^l = x^{cand}$  ▷ Accept the candidate
9: end for
```

Importance sampling

- Approximates integrals w/ form $E[f(x)] = \int f(x)P(x)$
- Choose density $Q(x)$
 - Should be close to $P(x)$
 - Assume: if $P(x) > 0$, then $Q(x) > 0$
- Generate r samples from $Q(x)$
 - Points where $Q(x) > P(x)$ will be overrepresented (and vice versa)
 - Introduce weights to account for this
 - $\hat{\Phi}$ converges to $E[x]$



$$\hat{\Phi} \equiv \frac{\sum_r w_r f(x^{(r)})}{\sum_r w_r}$$

$$w_r \equiv \frac{P^*(x^{(r)})}{Q^*(x^{(r)})}$$

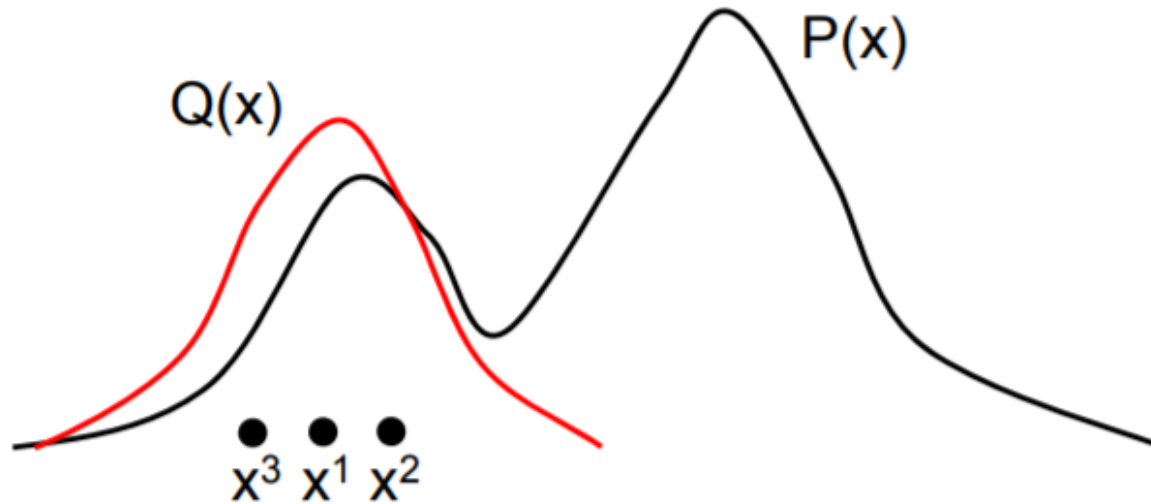
Shortcomings

- Variance of I.S. expectation is hard to estimate
- If $Q(x)$ is small where $P^*(x)$ is large, then it's quite possible that no samples fall in that region
- Both rejection sampling and importance sampling fail in high dimensions!
- Both require good proposal (close to $P(x)$) distributions, but if we knew the density $P(x)$ we would simply sample from it

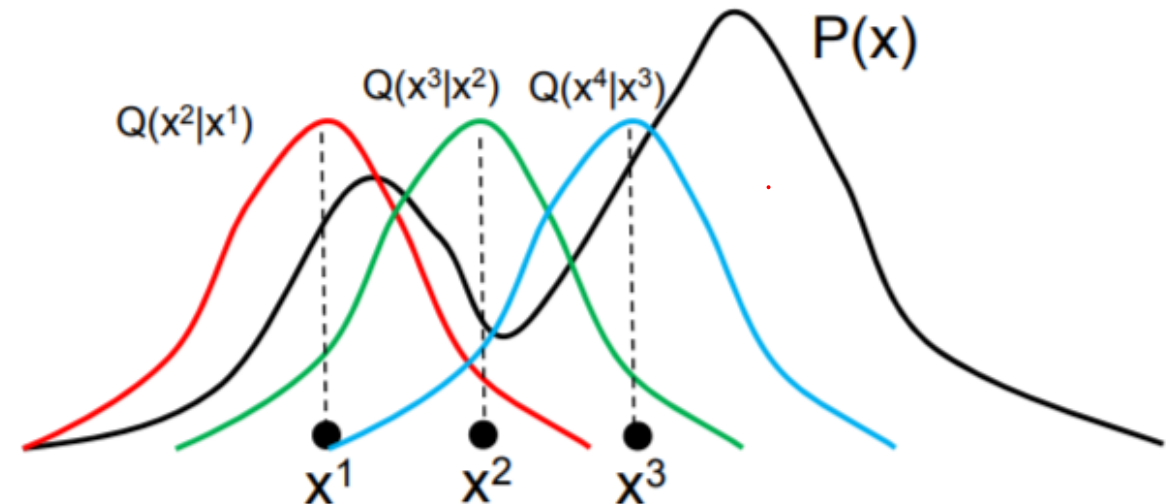
Adaptive proposals

- MCMC methods use an “adaptive proposal”
 - Instead of $Q(x)$, they use $Q(x^t|x^{t-1})$

Importance sampling with
a (bad) proposal $Q(x)$



MCMC with adaptive
proposal $Q(x'|x)$



Markov chain Monte Carlo

- A survey at the turn of the century put the Metropolis algorithm as one of the 10 most influential algorithms of the 20th century¹.
- The Metropolis algorithm is one of many sampling-type algorithms known as Markov chain Monte Carlo (MCMC).

[¹ J. Dongarra ; F. Sullivan, Computing in Science & Engineering, 2000]

Markov chain Monte Carlo

- MCMC methods are general algorithms for sampling from high-dimensional, complex distributions
- What are some of the more difficult to compute probability distributions?

posterior distribution

$$P(\pi|\mathcal{X}) = \frac{P(\mathcal{X}|\pi)P(\pi)}{P(\mathcal{X})} = \frac{P(\mathcal{X}|\pi)P(\pi)}{\int_{\pi} P(\mathcal{X}|\pi)P(\pi) d\pi}$$

posterior predictive distribution

$$\begin{aligned} p(x_{n+1} | x_{1:n}) &= \int p(x_{n+1}, \pi | x_{1:n}) d\pi \\ &= \int p(\pi | x_{1:n}) p(x_{n+1} | \pi) d\pi \\ &= \mathbb{E} [p(x_{n+1} | \pi) | x_{1:n}]. \end{aligned}$$

Bayes estimate

$$\hat{\pi}_{Bayes}(x) = \mathbb{E} [\pi | x]$$

Approximating integrals

- Recall conditional expectation $E[f(z)] = \int f(z)p(z) \, dz$
- Markov chains define a way to properly sample from $p(z)$ to compute expectations of this form
- Intuition: sum up $f(z)p(z)$ over infinitely many values of z
- If we sample a series $z^0, z^1, \dots, z^N$ **from probability density $p(z)$** then,

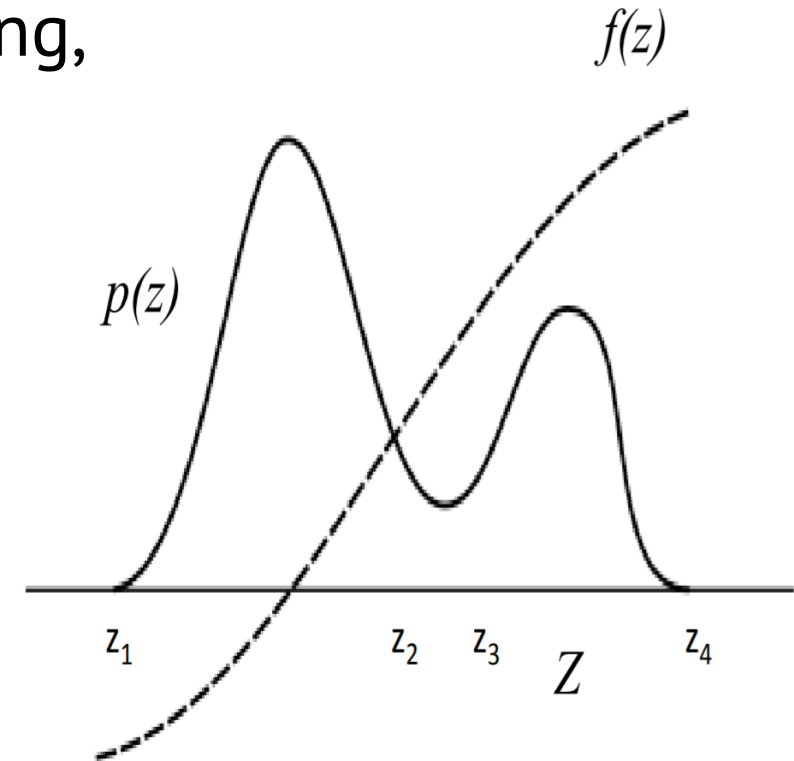
$$E_{p(z)}[f(z)] = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{t=1}^N f(z^{(t)})$$

How do we sample $z^0, z^1, \dots, z^N$?

- Many ways, importance sampling, rejection sampling, etc...
- What is important is that the sampling strategy samples z proportional to $p(z)$

Simple sampling strategy

- 1: $z^{(0)} :=$ a random initial point
- 2: **for** $t = 1$ to T **do**
- 3: $z^{(t+1)} := g(z^{(t)})$
- 4: **end for**



The “Markov” assumption for g

$$P_{\text{trans}}(z^{(t+1)} | z^{(0)}, z^{(1)}, \dots, z^{(t)}) = P_{\text{trans}}(z^{(t+1)} | z^{(t)})$$

Gibbs sampling

- Assume Z has $k > 1$ dimensions
- Gibbs sampling walks through $p(z)$ by

```
1:  $z^{(0)} := \langle z_1^{(0)}, \dots, z_k^{(0)} \rangle$   
2: for  $t = 1$  to  $T$  do  
3:   for  $i = 1$  to  $k$  do  
4:      $z_i^{(t+1)} \sim P(Z_i | z_1^{(t+1)}, \dots, z_{i-1}^{(t+1)}, z_{i+1}^{(t)}, \dots, z_k^{(t)})$   
5:   end for  
6: end for
```

- What is the update in (4)?

Gibbs sampling

- Assume Z has $k > 1$ dimensions
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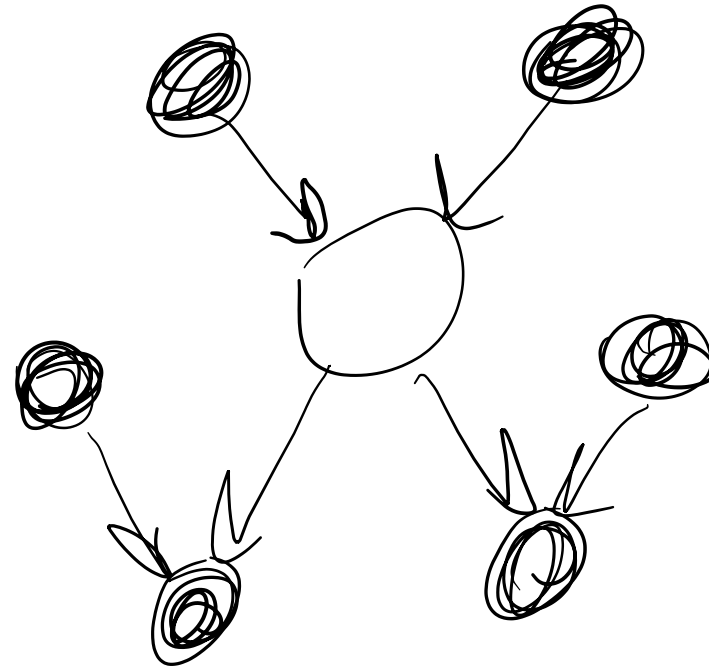
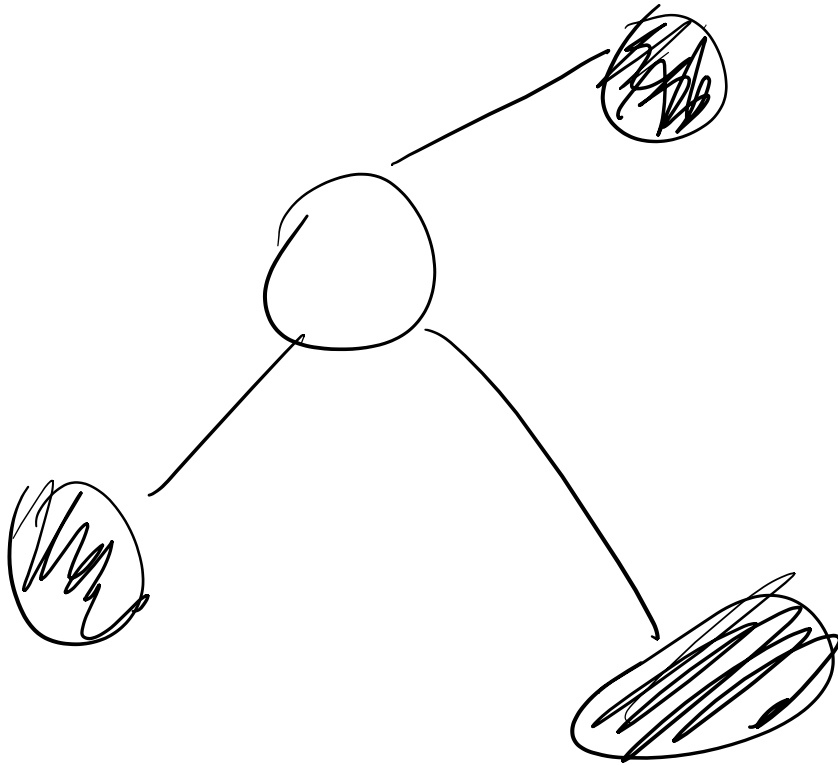
```
1:  $z^{(0)} := \langle z_1^{(0)}, \dots, z_k^{(0)} \rangle$ 
2: for  $t = 1$  to  $T$  do
3:   for  $i = 1$  to  $k$  do
4:      $z_i^{(t+1)} \sim P(Z_i | z_1^{(t+1)}, \dots, z_{i-1}^{(t+1)}, z_{i+1}^{(t)}, \dots, z_k^{(t)})$ 
5:   end for
6: end for
```

- What is the update in (4)?

$$P(Z_i | z_1^{(t+1)}, \dots, z_{i-1}^{(t+1)}, z_{i+1}^{(t)}, \dots, z_k^{(t)}) = \frac{P(z_1^{(t+1)}, \dots, z_{i-1}^{(t+1)}, z_i^{(t)}, z_{i+1}^{(t)}, \dots, z_k^{(t)})}{P(z_1^{(t+1)}, \dots, z_{i-1}^{(t+1)}, z_{i+1}^{(t)}, \dots, z_k^{(t)})}$$

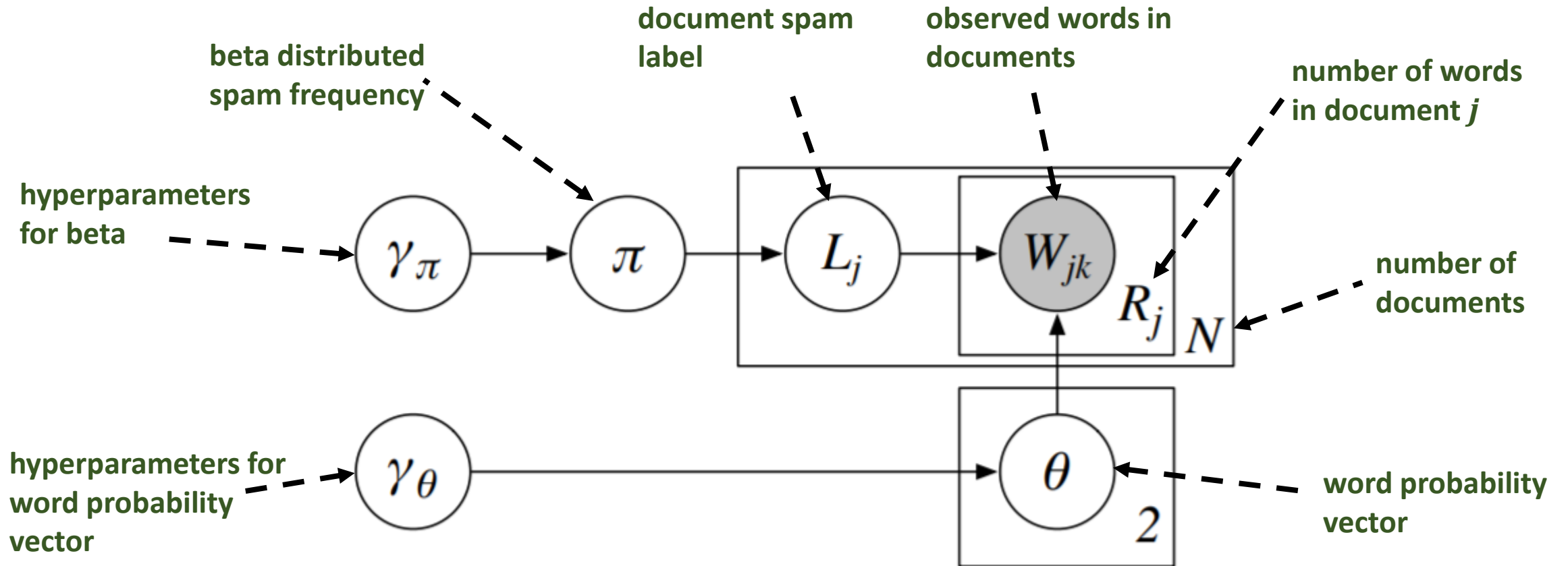
Markov blanket

- Difficulty depends on how hard it is to sample from complete conditionals
- Complete conditionals defined by a node's Markov blanket

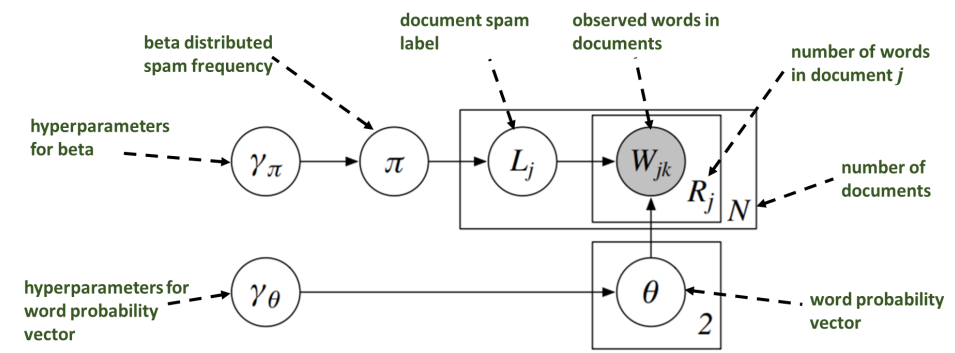


A Gibbs Sampler derivation

- Consider our Naïve Bayes model for spam classification



Generative model



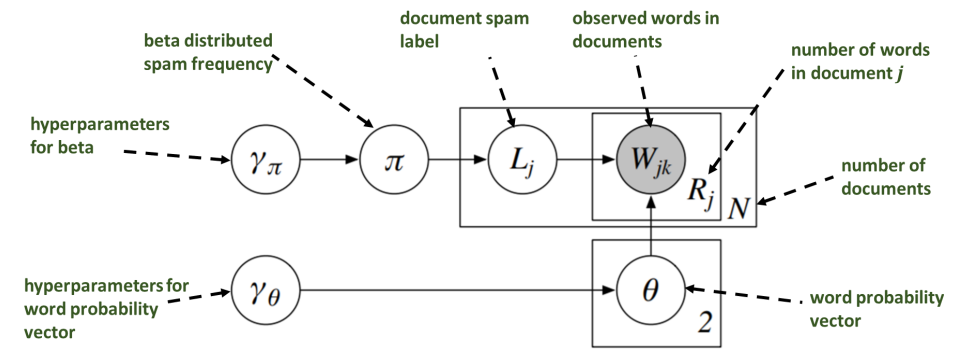
- Start from unconditioned variables and work our way in
- We assume documents are labelled as spam or not spam, this is a discrete probability with two classes

$$\pi \sim \text{Beta}(\gamma_\pi)$$

- Each document's label (spam or not spam) is then generated using π

$$L_j \sim \text{Bernoulli}(\pi).$$

Generative model



- There are two V -dimensional probability vectors for word distributions

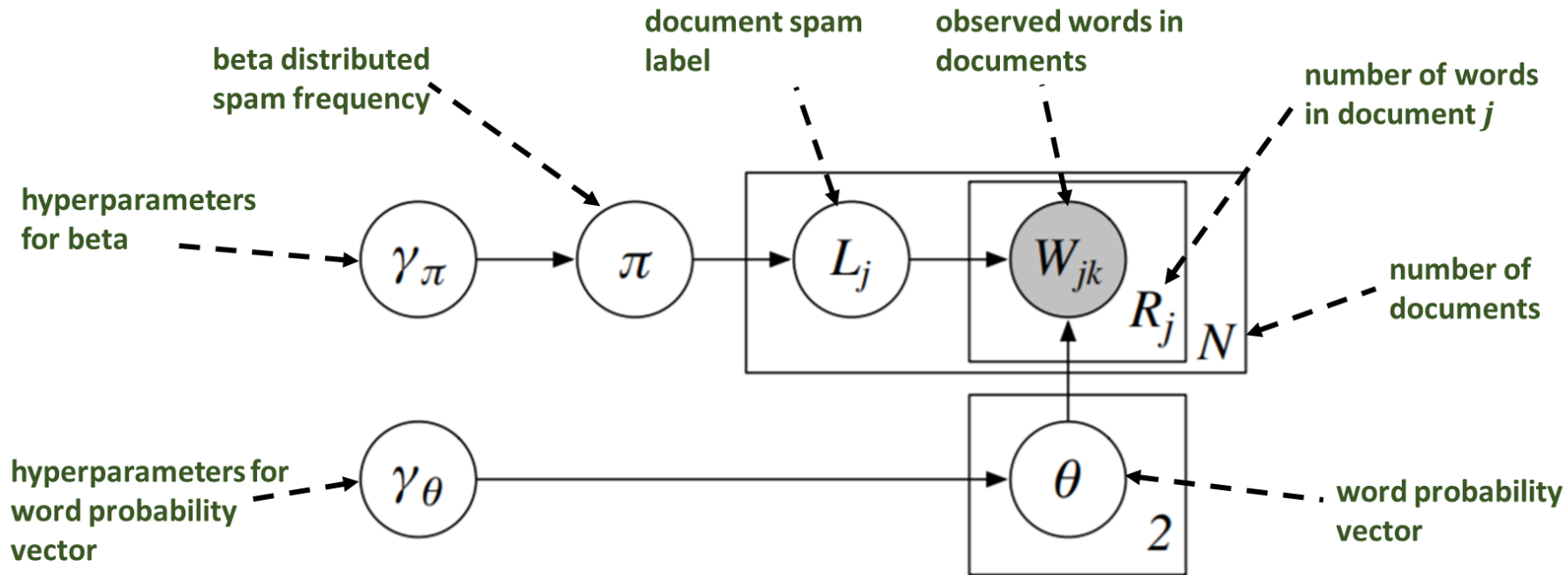
$$\boldsymbol{\theta} \sim \text{Dirichlet}(\boldsymbol{\gamma}_\theta)$$

- Each document is generated according to

$$\mathbf{W}_j \sim \text{Multinomial}(R_j, \boldsymbol{\theta}_{L_j})$$

State space

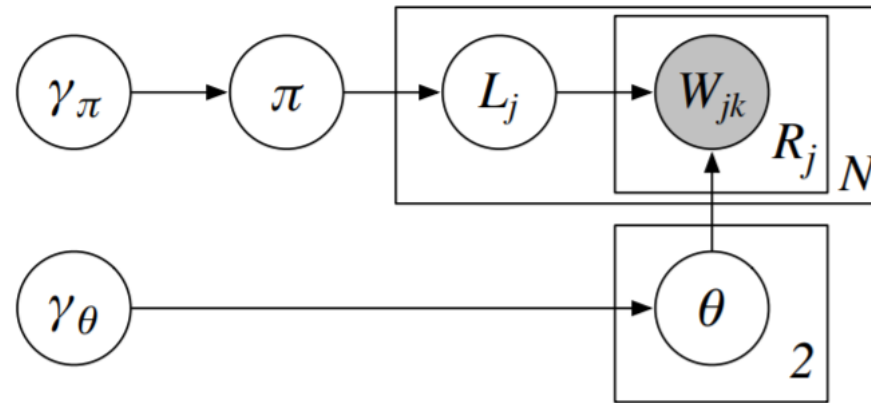
- In order to consider the joint, we write out the latent state space
 - π – scalar valued variable
 - θ_1, θ_2 – vector valued variables
 - $L - N$ binary valued variables



Joint distribution

$$P(\mathbb{C}, \mathbf{L}, \pi, \boldsymbol{\theta}_0, \boldsymbol{\theta}_1; \gamma_{\pi 1}, \gamma_{\pi 0}, \boldsymbol{\gamma}_{\theta})$$

$\mathbb{C}_x$ = set of documents labeled x
 $\mathbb{C}$ = set of all documents
 C_0 = number of documents labeled 0
 $N_{\mathbb{C}_x}(i)$ = number of times word i
 appears in documents $\mathbb{C}_x$



What is the Green unnormalized density?

What is the Blue unnormalized density?

$$P(\mathbb{C}, \mathbf{L}, \pi, \boldsymbol{\theta}_0, \boldsymbol{\theta}_1; \boldsymbol{\mu}) \propto \pi^{C_1 + \gamma_{\pi 1} - 1} (1 - \pi)^{C_0 + \gamma_{\pi 0} - 1} \prod_{i=1}^V \theta_{0,i}^{\mathcal{N}_{\mathbb{C}_0}(i) + \gamma_{\theta i} - 1} \theta_{1,i}^{\mathcal{N}_{\mathbb{C}_1}(i) + \gamma_{\theta i} - 1}$$

“Collapsed” Gibbs Sampler

- Collapsing means integrating out a subset of parameters
- The variable still affects the joint, but we don't need to explicitly update it

$$P(\mathbf{L}, \mathbb{C}, \boldsymbol{\theta}_0, \boldsymbol{\theta}_1; \boldsymbol{\mu}) = \int_{\pi} P(\mathbf{L}, \mathbb{C}, \boldsymbol{\theta}_0, \boldsymbol{\theta}_1, \pi; \boldsymbol{\mu}) \, d\pi$$

$$= \int_{\pi} P(\pi | \gamma_{\pi 1}, \gamma_{\pi 0}) P(\mathbf{L} | \pi) P(\boldsymbol{\theta}_0 | \gamma_{\theta}) P(\boldsymbol{\theta}_1 | \gamma_{\theta}) P(\mathbb{C}_0 | \boldsymbol{\theta}_0, \mathbf{L}) P(\mathbb{C}_1 | \boldsymbol{\theta}_1, \mathbf{L}) \, d\pi$$

$$= P(\boldsymbol{\theta}_0 | \gamma_{\theta}) P(\boldsymbol{\theta}_1 | \gamma_{\theta}) P(\mathbb{C}_0 | \boldsymbol{\theta}_0, \mathbf{L}) P(\mathbb{C}_1 | \boldsymbol{\theta}_1, \mathbf{L}) \int_{\pi} P(\pi | \gamma_{\pi 1}, \gamma_{\pi 0}) P(\mathbf{L} | \pi) \, d\pi$$

“Collapsed” Gibbs Sampler

$$\begin{aligned}\int_{\pi} P(\pi|\gamma_{\pi 1}, \gamma_{\pi 0}) P(\mathbf{L}|\pi) \, d\pi &= \int_{\pi} \frac{\Gamma(\gamma_{\pi 1} + \gamma_{\pi 0})}{\Gamma(\gamma_{\pi 1})\Gamma(\gamma_{\pi 0})} \pi^{\gamma_{\pi 1}-1} (1-\pi)^{\gamma_{\pi 0}-1} \pi^{C_1} (1-\pi)^{C_0} \, d\pi \\ &= \frac{\Gamma(\gamma_{\pi 1} + \gamma_{\pi 0})}{\Gamma(\gamma_{\pi 1})\Gamma(\gamma_{\pi 0})} \int_{\pi} \pi^{C_1+\gamma_{\pi 1}-1} (1-\pi)^{C_0+\gamma_{\pi 0}-1} \, d\pi\end{aligned}$$

- How do we evaluate this integral? What is the integrand?

“Collapsed” Gibbs Sampler

$$\begin{aligned}\int_{\pi} P(\pi|\gamma_{\pi 1}, \gamma_{\pi 0}) P(\mathbf{L}|\pi) \, d\pi &= \int_{\pi} \frac{\Gamma(\gamma_{\pi 1} + \gamma_{\pi 0})}{\Gamma(\gamma_{\pi 1})\Gamma(\gamma_{\pi 0})} \pi^{\gamma_{\pi 1}-1} (1-\pi)^{\gamma_{\pi 0}-1} \pi^{C_1} (1-\pi)^{C_0} \, d\pi \\ &= \frac{\Gamma(\gamma_{\pi 1} + \gamma_{\pi 0})}{\Gamma(\gamma_{\pi 1})\Gamma(\gamma_{\pi 0})} \int_{\pi} \pi^{C_1+\gamma_{\pi 1}-1} (1-\pi)^{C_0+\gamma_{\pi 0}-1} \, d\pi\end{aligned}$$

- But the integrand is simply an unnormalized beta distribution with $\alpha = C_1 + \gamma_{\pi 1}$ and $\beta = C_0 + \gamma_{\pi 0}$

$$\int_{\pi} \pi^{C_1+\gamma_{\pi 1}-1} (1-\pi)^{C_0+\gamma_{\pi 0}-1} \, d\pi = \frac{\Gamma(C_1 + \gamma_{\pi 1})\Gamma(C_0 + \gamma_{\pi 0})}{\Gamma(C_0 + C_1 + \gamma_{\pi 1} + \gamma_{\pi 0})}$$

Sampling document labels

- Complete conditional for L_j (label for document j)

$$\begin{aligned} P(L_j | \mathbf{L}^{(-j)}, \mathbb{C}^{(-j)}, \boldsymbol{\theta}_0, \boldsymbol{\theta}_1; \boldsymbol{\mu}) &= \frac{P(L_j, \mathbf{W}_j, \mathbf{L}^{(-j)}, \mathbb{C}^{(-j)}, \boldsymbol{\theta}_0, \boldsymbol{\theta}_1; \boldsymbol{\mu})}{P(\mathbf{L}^{(-j)}, \mathbb{C}^{(-j)}, \boldsymbol{\theta}_0, \boldsymbol{\theta}_1; \boldsymbol{\mu})} \\ &= \frac{P(\mathbf{L}, \mathbb{C}, \boldsymbol{\theta}_0, \boldsymbol{\theta}_1; \boldsymbol{\mu})}{P(\mathbf{L}^{(-j)}, \mathbb{C}^{(-j)}, \boldsymbol{\theta}_0, \boldsymbol{\theta}_1; \boldsymbol{\mu})} \end{aligned}$$

- We saw the numerator when deriving the collapsed Gibbs sampler

$$P(\mathbf{L}, \mathbb{C}, \boldsymbol{\theta}_0, \boldsymbol{\theta}_1; \boldsymbol{\mu}) \propto \frac{\Gamma(\gamma_{\pi_1} + \gamma_{\pi_0})}{\Gamma(\gamma_{\pi_1})\Gamma(\gamma_{\pi_0})} \frac{\Gamma(C_1 + \gamma_{\pi_1})\Gamma(C_0 + \gamma_{\pi_0})}{\Gamma(N + \gamma_{\pi_1} + \gamma_{\pi_0})} \prod_{i=1}^V \theta_{0,i}^{\mathcal{N}_{\mathbb{C}_0}(i) + \gamma_{\theta_i} - 1} \theta_{1,i}^{\mathcal{N}_{\mathbb{C}_1}(i) + \gamma_{\theta_i} - 1}$$

- Take each term at a time...

Sampling document labels

$$\frac{P(\mathbf{L}, \mathbb{C}, \boldsymbol{\theta}_0, \boldsymbol{\theta}_1; \boldsymbol{\mu})}{P(\mathbf{L}^{(-j)}, \mathbb{C}^{(-j)}, \boldsymbol{\theta}_0, \boldsymbol{\theta}_1; \boldsymbol{\mu})}$$

$$P(\mathbf{L}, \mathbb{C}, \boldsymbol{\theta}_0, \boldsymbol{\theta}_1; \boldsymbol{\mu}) \propto \frac{\Gamma(\gamma_{\pi 1} + \gamma_{\pi 0})}{\Gamma(\gamma_{\pi 1})\Gamma(\gamma_{\pi 0})} \frac{\Gamma(C_1 + \gamma_{\pi 1})\Gamma(C_0 + \gamma_{\pi 0})}{\Gamma(N + \gamma_{\pi 1} + \gamma_{\pi 0})} \prod_{i=1}^V \theta_{0,i}^{\mathcal{N}_{\mathbb{C}_0}(i) + \gamma_{\theta i} - 1} \theta_{1,i}^{\mathcal{N}_{\mathbb{C}_1}(i) + \gamma_{\theta i} - 1}$$

No terms depend
on $\mathbf{L}$ or $\mathbb{C}$

By removing j , either C_1
or C_0 loses a document,
but not both.

$$C_x^{(-j)} = C_x - 1$$



$$\frac{\Gamma(C_x + \gamma_{\pi x})\Gamma(N + \gamma_{\pi 1} + \gamma_{\pi 0} - 1)}{\Gamma(N + \gamma_{\pi 1} + \gamma_{\pi 0})\Gamma(C_x + \gamma_{\pi x} - 1)}$$

Sampling document labels

$$\frac{P(\mathbf{L}, \mathbb{C}, \boldsymbol{\theta}_0, \boldsymbol{\theta}_1; \boldsymbol{\mu})}{P(\mathbf{L}^{(-j)}, \mathbb{C}^{(-j)}, \boldsymbol{\theta}_0, \boldsymbol{\theta}_1; \boldsymbol{\mu})}$$

$$P(\mathbf{L}, \mathbb{C}, \boldsymbol{\theta}_0, \boldsymbol{\theta}_1; \boldsymbol{\mu}) \propto \frac{\Gamma(\gamma_{\pi 1} + \gamma_{\pi 0})}{\Gamma(\gamma_{\pi 1})\Gamma(\gamma_{\pi 0})} \frac{\Gamma(C_1 + \gamma_{\pi 1})\Gamma(C_0 + \gamma_{\pi 0})}{\Gamma(N + \gamma_{\pi 1} + \gamma_{\pi 0})} \prod_{i=1}^V \theta_{0,i}^{\mathcal{N}_{\mathbb{C}_0}(i) + \gamma_{\theta i} - 1} \theta_{1,i}^{\mathcal{N}_{\mathbb{C}_1}(i) + \gamma_{\theta i} - 1}$$

No terms depend
on $\mathbf{L}$ or $\mathbb{C}$

By removing j , either C_1
or C_0 loses a document,
but not both.

By removing j , again,
one class is changed

$$C_x^{(-j)} = C_x - 1$$

↓

$$\frac{\Gamma(C_x + \gamma_{\pi x})\Gamma(N + \gamma_{\pi 1} + \gamma_{\pi 0} - 1)}{\Gamma(N + \gamma_{\pi 1} + \gamma_{\pi 0})\Gamma(C_x + \gamma_{\pi x} - 1)} = \frac{C_x + \gamma_{\pi x} - 1}{N + \gamma_{\pi 1} + \gamma_{\pi 0} - 1}$$

$$\Gamma(N + \gamma_{\pi 1} + \gamma_{\pi 0}) = (N + \gamma_{\pi 1} + \gamma_{\pi 0} - 1)\Gamma(N + \gamma_{\pi 1} + \gamma_{\pi 0} - 1)$$

$$\Gamma(C_x + \gamma_{\pi x}) = (C_x + \gamma_{\pi x} - 1)\Gamma(C_x + \gamma_{\pi x} - 1)$$

Sampling document labels

$$\frac{P(\mathbf{L}, \mathbb{C}, \boldsymbol{\theta}_0, \boldsymbol{\theta}_1; \boldsymbol{\mu})}{P(\mathbf{L}^{(-j)}, \mathbb{C}^{(-j)}, \boldsymbol{\theta}_0, \boldsymbol{\theta}_1; \boldsymbol{\mu})}$$

$$P(\mathbf{L}, \mathbb{C}, \boldsymbol{\theta}_0, \boldsymbol{\theta}_1; \boldsymbol{\mu}) \propto \underbrace{\frac{\Gamma(\gamma_{\pi 1} + \gamma_{\pi 0})}{\Gamma(\gamma_{\pi 1})\Gamma(\gamma_{\pi 0})}}_{\text{No terms depend on } \mathbf{L} \text{ or } \mathbb{C}} \underbrace{\frac{\Gamma(C_1 + \gamma_{\pi 1})\Gamma(C_0 + \gamma_{\pi 0})}{\Gamma(N + \gamma_{\pi 1} + \gamma_{\pi 0})}}_{\text{By removing } j, \text{ either } C_1 \text{ or } C_0 \text{ loses a document, but not both.}} \underbrace{\prod_{i=1}^V \theta_{0,i}^{\mathcal{N}_{\mathbb{C}_0}(i) + \gamma_{\theta i} - 1} \theta_{1,i}^{\mathcal{N}_{\mathbb{C}_1}(i) + \gamma_{\theta i} - 1}}_{\text{By removing } j, \text{ again, one class is changed}}$$

No terms depend
on $\mathbf{L}$ or $\mathbb{C}$

By removing j , either C_1
or C_0 loses a document,
but not both.

By removing j , again,
one class is changed



$$\prod_{i=1}^V \frac{\theta_{x,i}^{\mathcal{N}_{\mathbb{C}_x}(i) + \gamma_{\theta i} - 1}}{\theta_{x,i}^{\mathcal{N}_{\mathbb{C}_x^{(-j)}}(i) + \gamma_{\theta i} - 1}} = \prod_{i=1}^V \theta_{x,i}^{W_{ji}}$$

Sampling document labels (interpretation)

$$\Pr(\mathbf{L}_j = x | \mathbf{L}^{(-j)}, \mathbb{C}^{(-j)}, \boldsymbol{\theta}_0, \boldsymbol{\theta}_1; \boldsymbol{\mu}) = \frac{C_x + \gamma_{\pi x} - 1}{N + \gamma_{\pi 1} + \gamma_{\pi 0} - 1} \prod_{i=1}^V \theta_{x,i}^{W_{ji}}$$

Sampling document labels (interpretation)

$$\Pr(\mathbf{L}_j = x | \mathbf{L}^{(-j)}, \mathbb{C}^{(-j)}, \boldsymbol{\theta}_0, \boldsymbol{\theta}_1; \boldsymbol{\mu}) = \frac{C_x + \gamma_{\pi x} - 1}{N + \gamma_{\pi 1} + \gamma_{\pi 0} - 1} \prod_{i=1}^V \theta_{x,i}^{W_{ji}}$$

Pushing document towards
class with higher prior or more
documents with label x

A likelihood
term

Sampling document labels (interpretation)

- Putting it all together
 - Compute $a = \Pr(L_j = 0 | L^{(-j)}, \mathbb{C}^{(-j)}, \boldsymbol{\theta}_0, \boldsymbol{\theta}_1; \mu)$
 - Compute $b = \Pr(L_j = 1 | L^{(-j)}, \mathbb{C}^{(-j)}, \boldsymbol{\theta}_0, \boldsymbol{\theta}_1; \mu)$
 - Sample a label $L_j^{(t+1)} \sim \text{Bern}\left(\frac{a}{a+b}\right)$
- Other update is easier, see Gamma distribution trick to sample from a Dirichlet
$$P(\boldsymbol{\theta} | \mathbb{C}, \mathbf{L}; \boldsymbol{\mu}) \propto P(\mathbb{C}, \mathbf{L} | \boldsymbol{\theta}) P(\boldsymbol{\theta} | \boldsymbol{\mu})$$
- Incorporate labels is easy, simply set their values and do not update their parameters

Full algorithm

```
1: for  $t := 1$  to  $T$  do
2:   for  $j := 1$  to  $N$  do
3:     if  $j$  is not a training document then
4:       Subtract  $j$ 's word counts from the total word counts of whatever class it's currently a member of
5:       Subtract 1 from the count of documents with label  $L_j$ 
6:       Assign a new label  $L_j^{(t+1)}$  to document  $j$  as described at the end of Section 2.5.1
7:       Add 1 to the count of documents with label  $L_j^{(t+1)}$ 
8:       Add  $j$ 's word counts to the total word counts for class  $L_j^{(t+1)}$ 
9:     end if
10:  end for
11:   $\mathbf{t}_0 :=$  vector of total word counts from class 0, including pseudocounts
12:   $\boldsymbol{\theta}_0 \sim \text{Dirichlet}(\mathbf{t}_0)$ , as described in Section 2.5.2
13:   $\mathbf{t}_1 :=$  vector of total word counts from class 1, including pseudocounts
14:   $\boldsymbol{\theta}_1 \sim \text{Dirichlet}(\mathbf{t}_1)$ , as described in Section 2.5.2
15: end for
```

Metropolis-Hastings

- Draw x' from $Q(x'|x)$ where x was the previous sample
- Accept x' with probability

- $A(x'|x)$ is the ratio of importance sampling weights

- $\frac{P(x')}{Q(x'|x)}$ importance weight for x'

$$A(x'|x) = \min\left(1, \frac{P(x')Q(x|x')}{P(x)Q(x'|x)}\right)$$

- $\frac{P(x)}{Q(x|x')}$ importance weight for x

The MH Algorithm

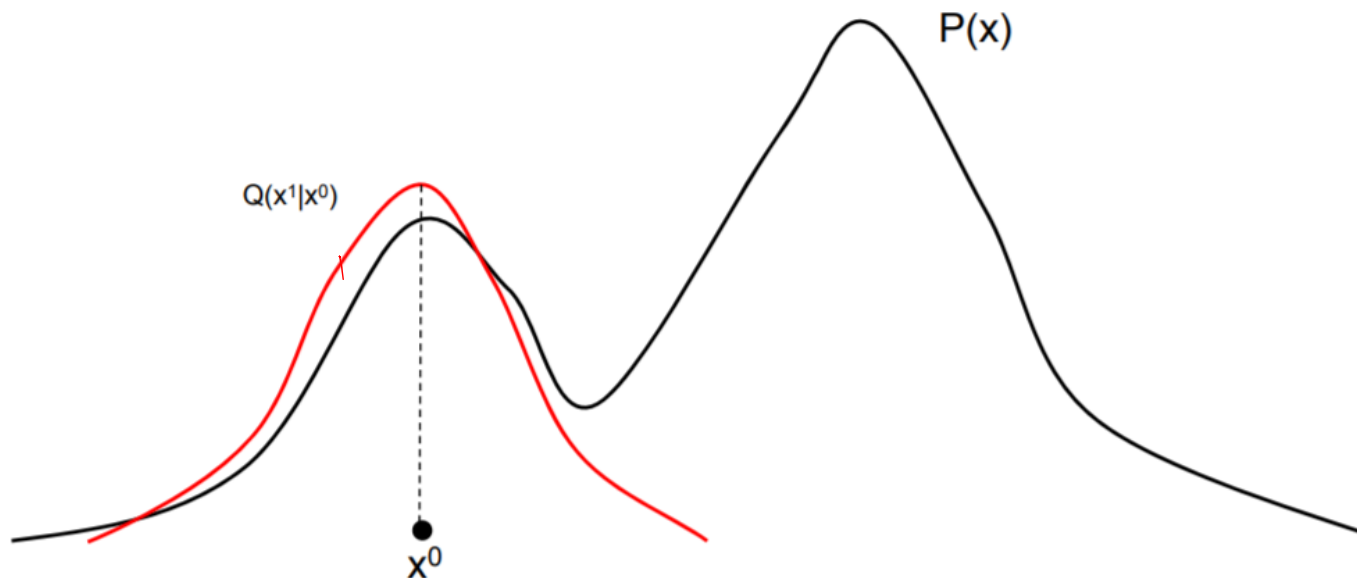
$$A(x'|x) = \min\left(1, \frac{P(x')Q(x|x')}{P(x)Q(x'|x)}\right)$$



- Example:
 - Let $Q(x'|x)$ be a Gaussian centered on x
 - We're trying to sample from a bimodal distribution $P(x)$

Initialize $x^{(0)}$

...



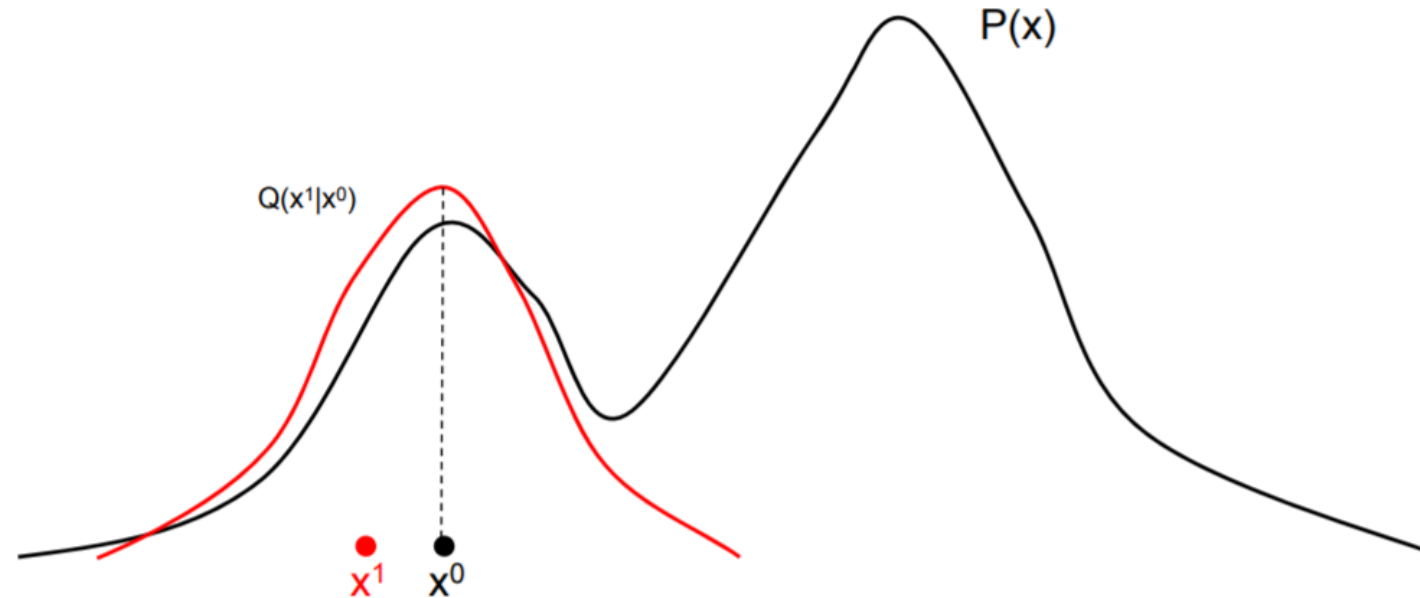
The MH Algorithm

$$A(x'|x) = \min\left(1, \frac{P(x')Q(x|x')}{P(x)Q(x'|x)}\right)$$



- Example:
 - Let $Q(x'|x)$ be a Gaussian centered on x
 - We're trying to sample from a bimodal distribution $P(x)$

Initialize $x^{(0)}$
Draw, accept x^1



The MH Algorithm

$$A(x'|x) = \min\left(1, \frac{P(x')Q(x|x')}{P(x)Q(x'|x)}\right)$$

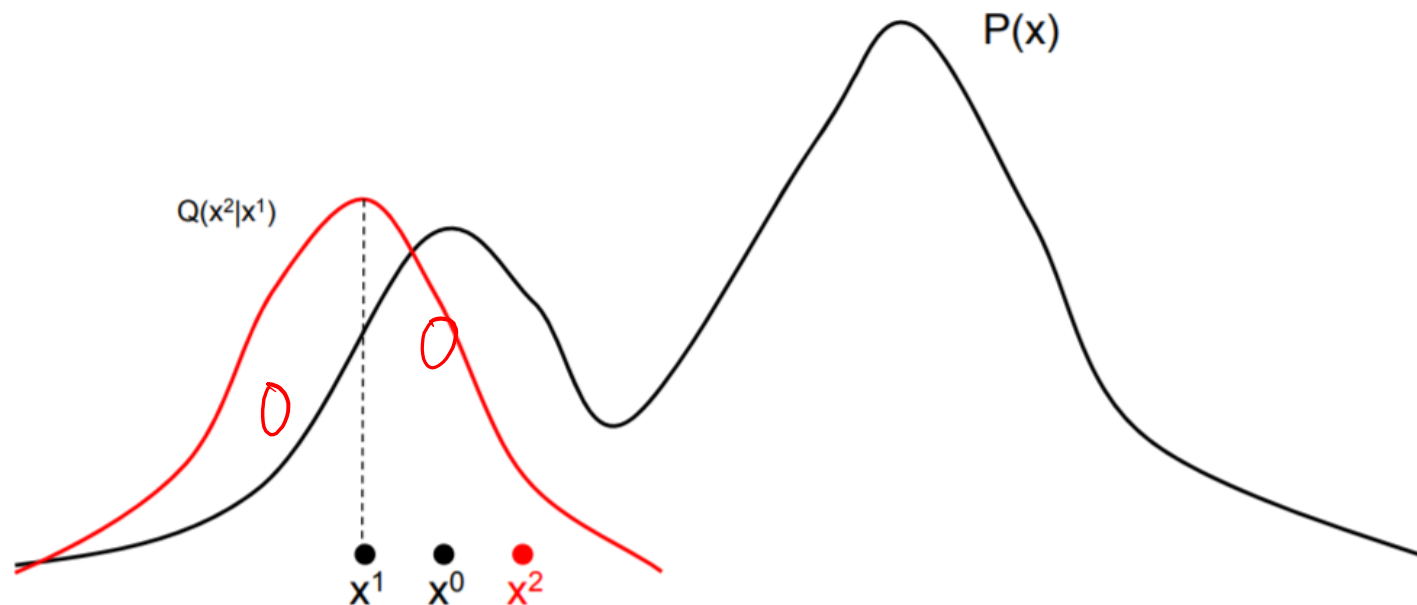


- Example:
 - Let $Q(x'|x)$ be a Gaussian centered on x
 - We're trying to sample from a bimodal distribution $P(x)$

Initialize $x^{(0)}$

Draw, accept x^1

Draw, accept x^2



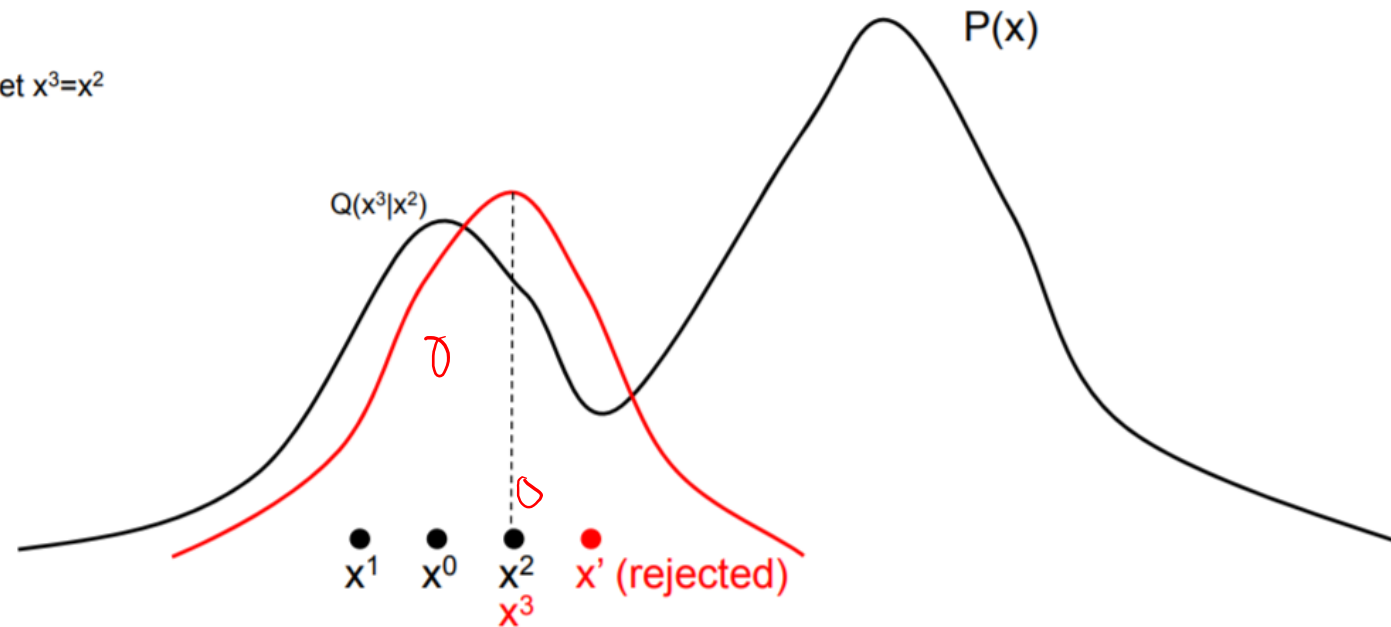
$$A(x'|x) = \min\left(1, \frac{P(x')Q(x|x')}{P(x)Q(x'|x)}\right)$$



The MH Algorithm

- Example:
 - Let $Q(x'|x)$ be a Gaussian centered on x
 - We're trying to sample from a bimodal distribution $P(x)$

Initialize $x^{(0)}$
Draw, accept x^1
Draw, accept x^2
Draw but reject; set $x^3=x^2$



The MH Algorithm

$$A(x'|x) = \min\left(1, \frac{P(x')Q(x|x')}{P(x)Q(x'|x)}\right)$$



- Example:
 - Let $Q(x'|x)$ be a Gaussian centered on x
 - We're trying to sample from a bimodal distribution $P(x)$

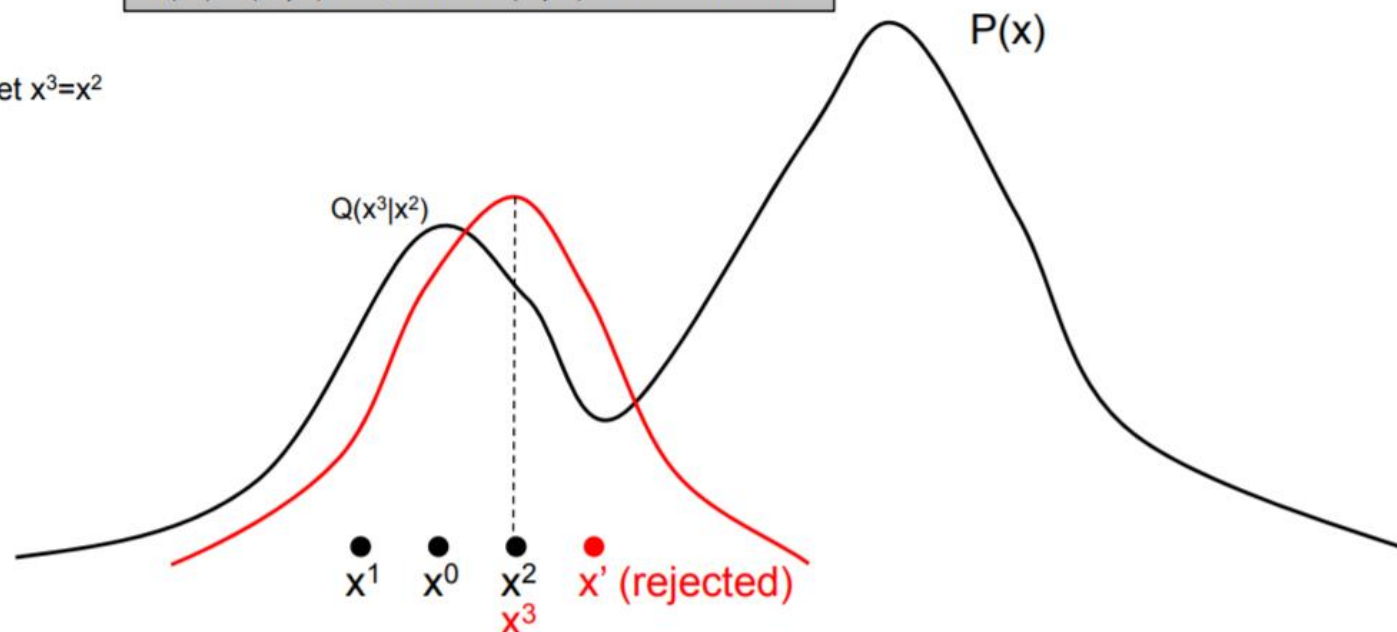
Initialize $x^{(0)}$

Draw, accept x^1

Draw, accept x^2

Draw but reject; set $x^3=x^2$

We reject because $P(x')/Q(x'|x^2) < 1$ and $P(x^2)/Q(x^2|x') > 1$, hence $A(x'|x^2)$ is close to zero!



The MH Algorithm

$$A(x'|x) = \min\left(1, \frac{P(x')Q(x|x')}{P(x)Q(x'|x)}\right)$$



- Example:
 - Let $Q(x'|x)$ be a Gaussian centered on x
 - We're trying to sample from a bimodal distribution $P(x)$

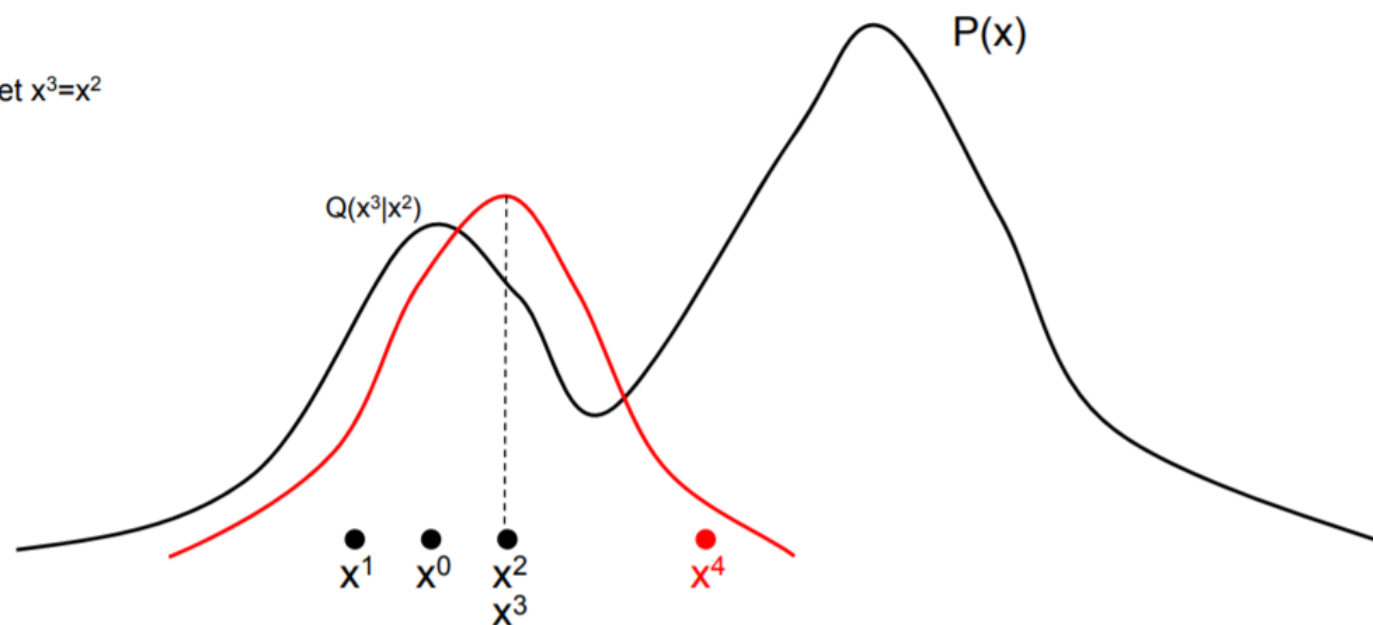
Initialize $x^{(0)}$

Draw, accept x^1

Draw, accept x^2

Draw but reject; set $x^3=x^2$

Draw, accept x^4



The MH Algorithm

$$A(x'|x) = \min\left(1, \frac{P(x')Q(x|x')}{P(x)Q(x'|x)}\right)$$



- Example:
 - Let $Q(x'|x)$ be a Gaussian centered on x
 - We're trying to sample from a bimodal distribution $P(x)$

Initialize $x^{(0)}$

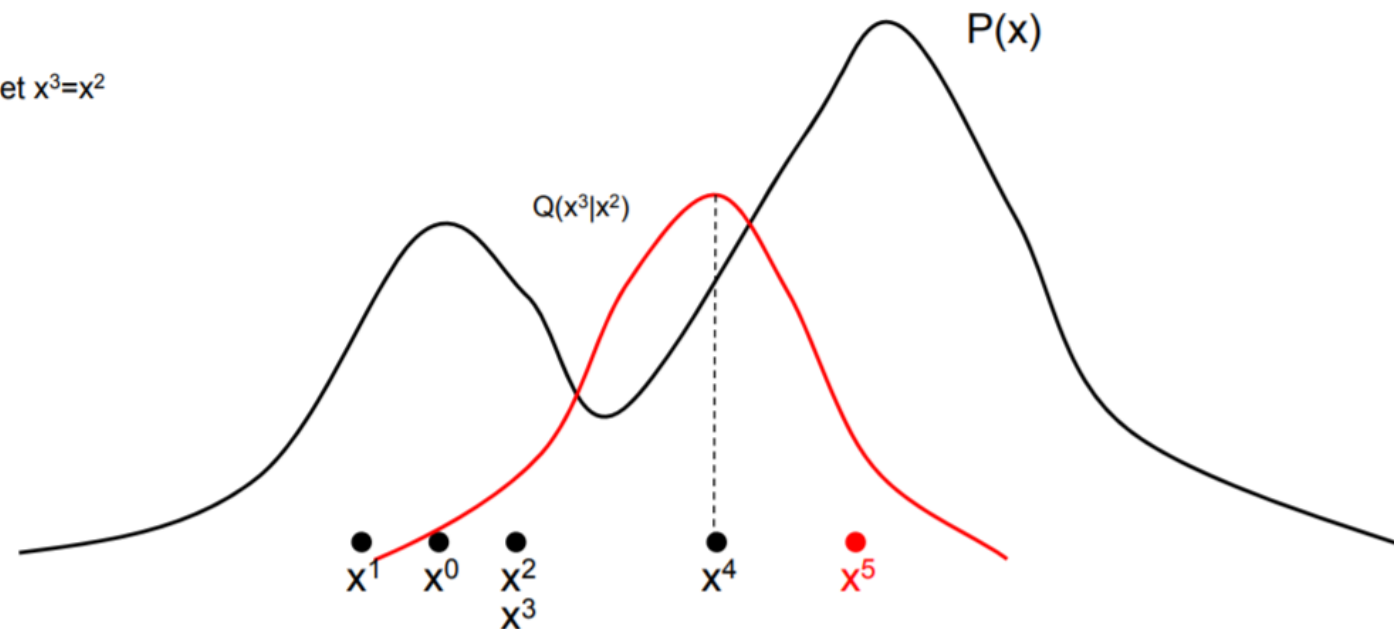
Draw, accept x^1

Draw, accept x^2

Draw but reject; set $x^3=x^2$

Draw, accept x^4

Draw, accept x^5



The MH Algorithm

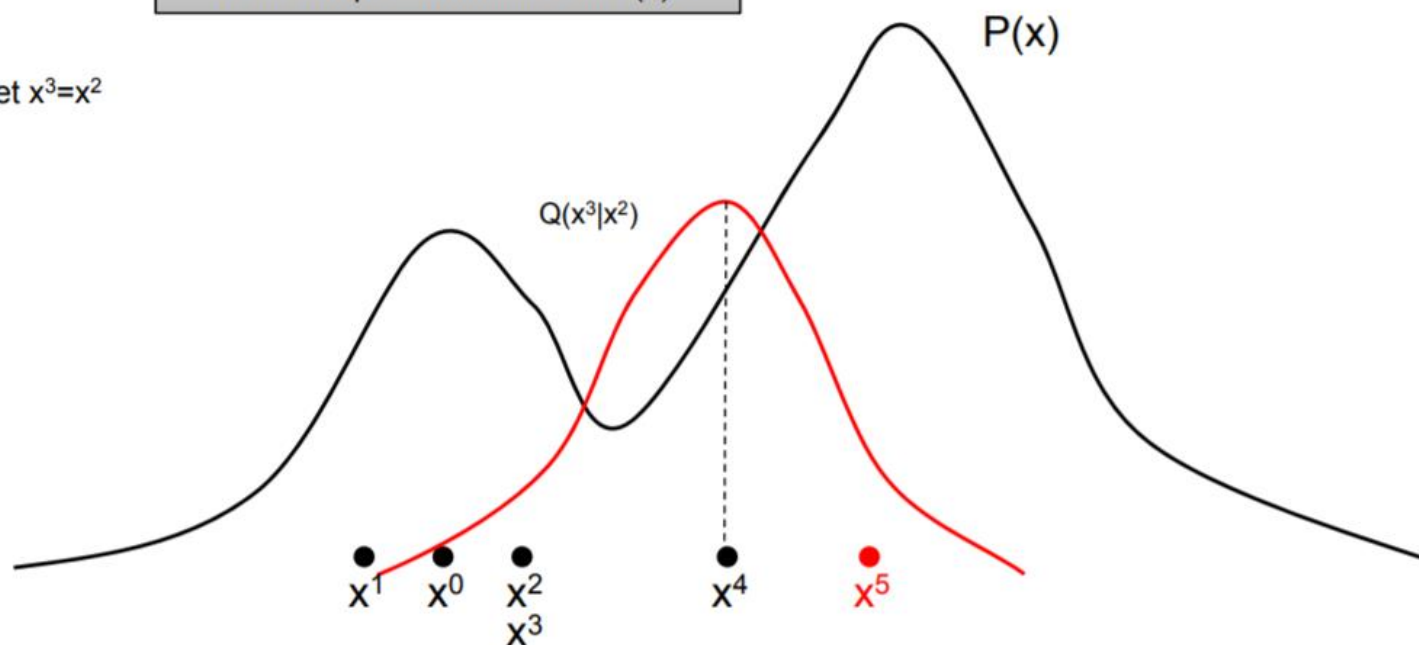
$$A(x' | x) = \min \left(1, \frac{P(x')Q(x | x')}{P(x)Q(x' | x)} \right)$$



- Example:
 - Let $Q(x'|x)$ be a Gaussian centered on x
 - We're trying to sample from a bimodal distribution $P(x)$

Initialize $x^{(0)}$
Draw, accept x^1
Draw, accept x^2
Draw but reject; set $x^3 = x^2$
Draw, accept x^4
Draw, accept x^5

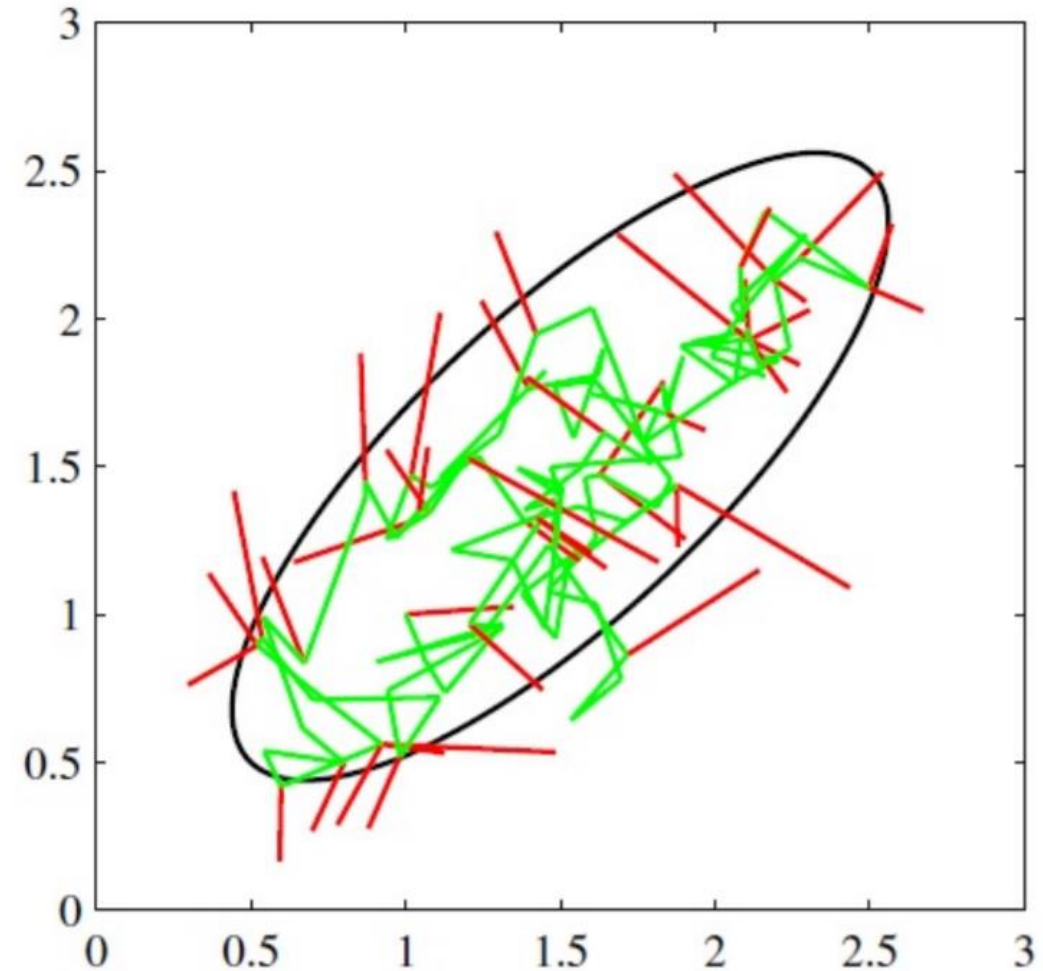
The adaptive proposal $Q(x'|x)$ allows us to sample both modes of $P(x)$!



MH algorithm

- 150 steps of MH on a Gaussian with proposal

$$Q(\cdot \mid \mathbf{x}_t) \sim \mathcal{N}_2(\mathbf{x}_t, 0.2 \times \mathbf{I}_2)$$



[\[Gibbs demo\]](#)

Gibbs Sampling is a special case of MH

- GS proposal distribution is

$$Q(x'_i, \mathbf{x}_{-i} | x_i, \mathbf{x}_{-i}) = P(x'_i | \mathbf{x}_{-i})$$

- Applying MH using this proposal we see that every proposal is accepted

$$\begin{aligned} A(x'_i, \mathbf{x}_{-i} | x_i, \mathbf{x}_{-i}) &= \min \left(1, \frac{P(x'_i, \mathbf{x}_{-i}) Q(x_i, \mathbf{x}_{-i} | x'_i, \mathbf{x}_{-i})}{P(x_i, \mathbf{x}_{-i}) Q(x'_i, \mathbf{x}_{-i} | x_i, \mathbf{x}_{-i})} \right) \\ &= \min \left(1, \frac{P(x'_i, \mathbf{x}_{-i}) P(x_i | \mathbf{x}_{-i})}{P(x_i, \mathbf{x}_{-i}) P(x'_i | \mathbf{x}_{-i})} \right) = \min \left(1, \frac{P(x'_i | \mathbf{x}_{-i}) P(\mathbf{x}_{-i}) P(x_i | \mathbf{x}_{-i})}{P(x_i | \mathbf{x}_{-i}) P(\mathbf{x}_{-i}) P(x'_i | \mathbf{x}_{-i})} \right) \\ &= \min(1, 1) = 1 \end{aligned}$$

Summary: A Gibbs Sampler derivation

- 1) Collapsing or blocking?
- 2) Write out complete conditionals
- 3) Compute update (only nodes in Markov blanket matter)
 - Analytical solution
 - Reduces to known distribution, reinstate constant
 - Discrete, normalize by sum
 - Continuous and unknown distribution, approximate or sample update...

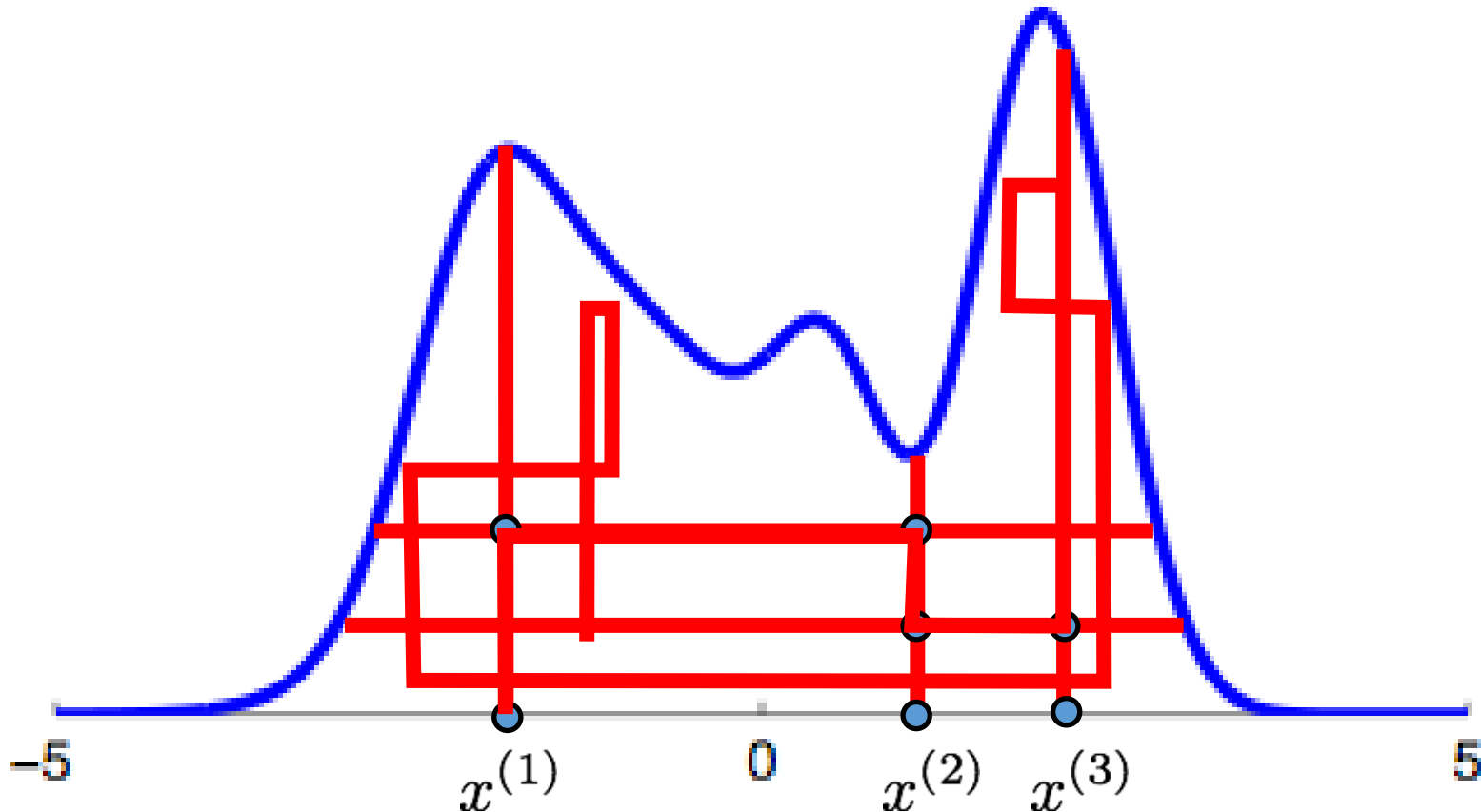
One more technique: slice sampling

- Draw samples from some function $P^*(x)$
- Transition from point to point lying under a curve

```
1: evaluate  $P^*(x)$ 
2: draw a vertical coordinate  $u' \sim \text{Uniform}(0, P^*(x))$ 
3: create a horizontal interval  $(x_l, x_r)$  enclosing  $x$ 
4: loop {
5:     draw  $x' \sim \text{Uniform}(x_l, x_r)$ 
6:     evaluate  $P^*(x')$ 
7:     if  $P^*(x') > u'$  break out of loop 4-9
8:     else modify the interval  $(x_l, x_r)$ 
9: }
```

Slice Sampling

Define a Markov chain that samples uniformly from the area beneath the curve. This means that we need to introduce a “height” into the MCMC sampler.

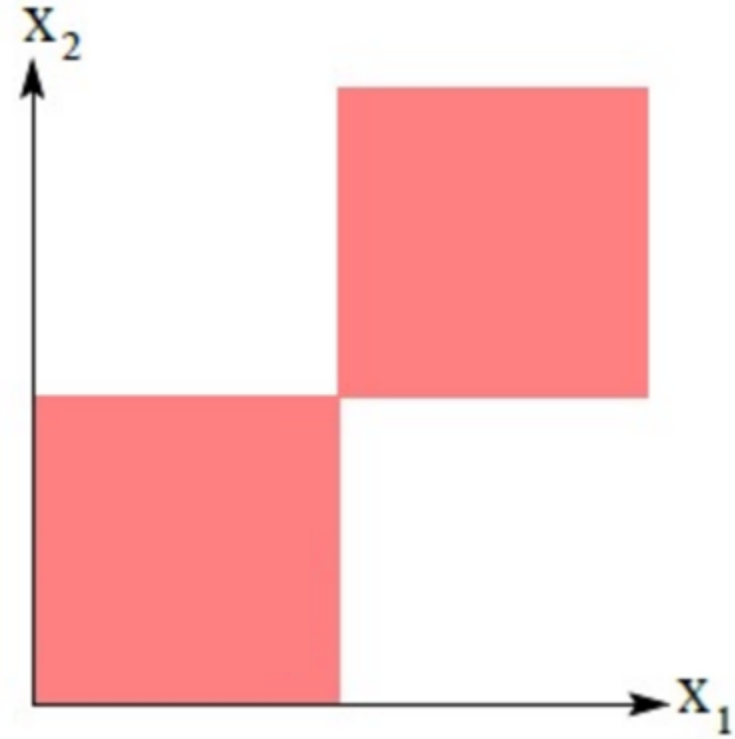


Why does this work? (in notes)

- MCMC defines a **MC** over latent variables that converges to the posterior
- **Irreducibility**: no state is unreachable

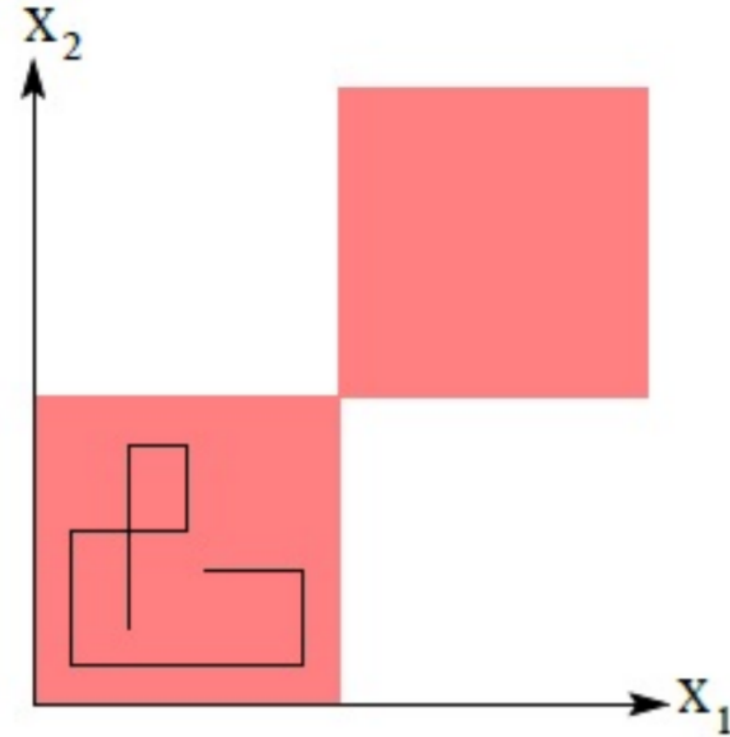
Markov chain for Gibbs Sampling

- Will a Gibbs sampler converge to $p(x_1, x_2)$ given the two dimensional distribution on the right?



Markov chain for Gibbs Sampling

- Will a Gibbs sampler converge to $p(x_1, x_2)$ given the two dimensional distribution on the right?
- No, $p(x|x') = 0$ for x in the upper right quadrant and x' in the lower left (and vice versa).

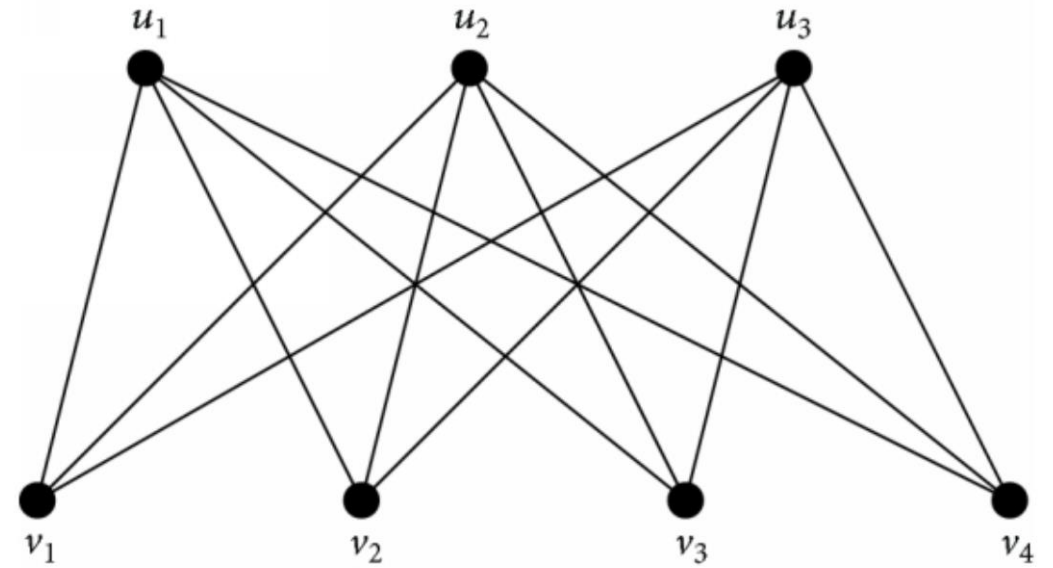


Why does this work? (in notes)

- MCMC defines a **MC** over latent variables that converges to the posterior
- **Irreducibility**: no state is unreachable
- **Aperiodicity**: no states require a specific number of steps to revisit

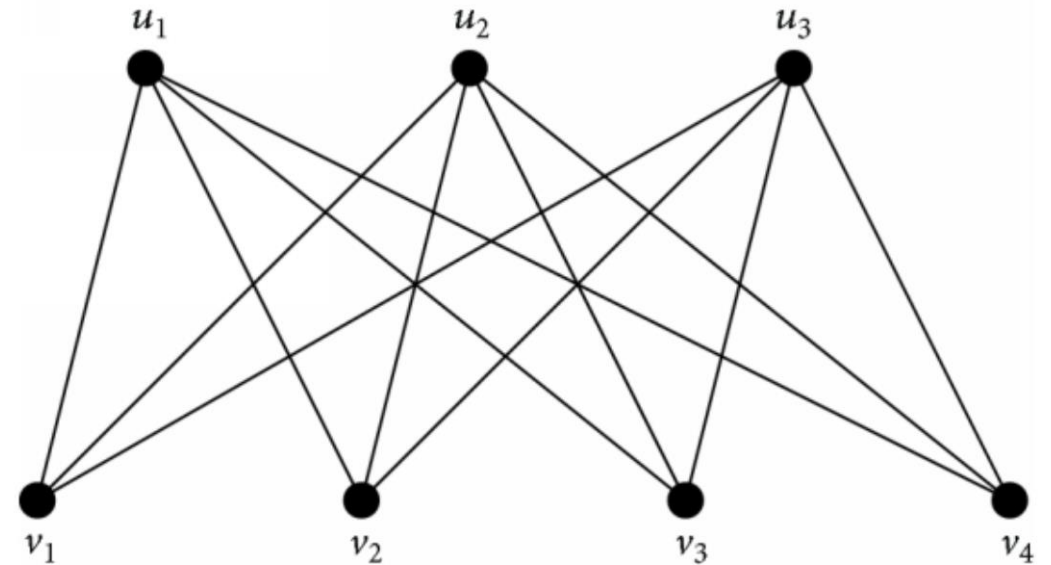
Markov chain for Gibbs Sampling

- Will a Gibbs sampler with Markov process defined on the right converge?



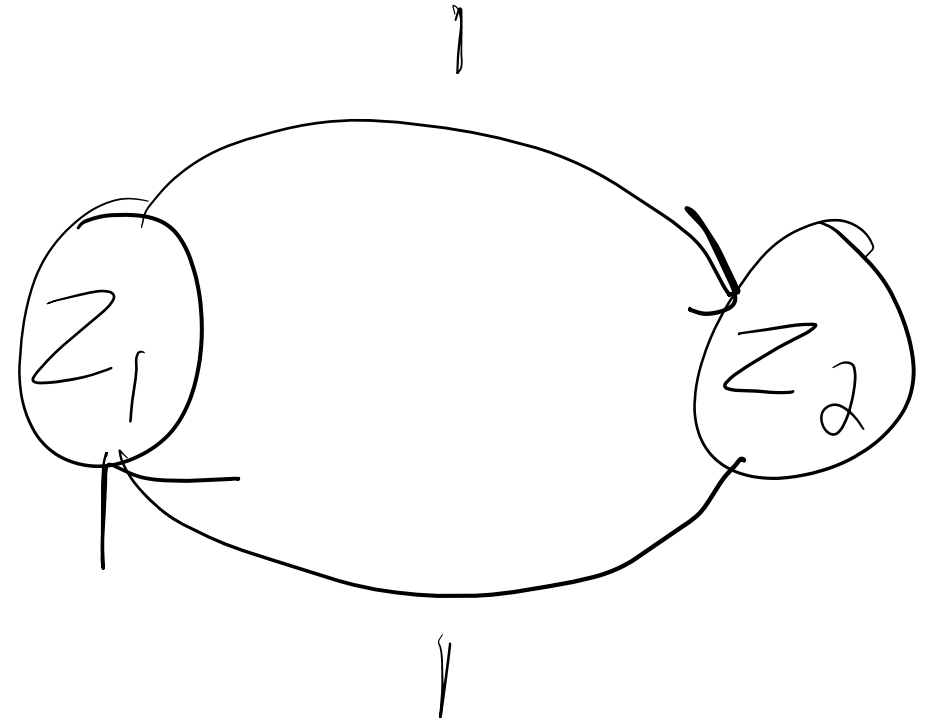
Markov chain for Gibbs Sampling

- Will a Gibbs sampler with Markov process defined on the right converge?
- No, bipartite graphs are cyclic with even period



Markov chain for Gibbs Sampling

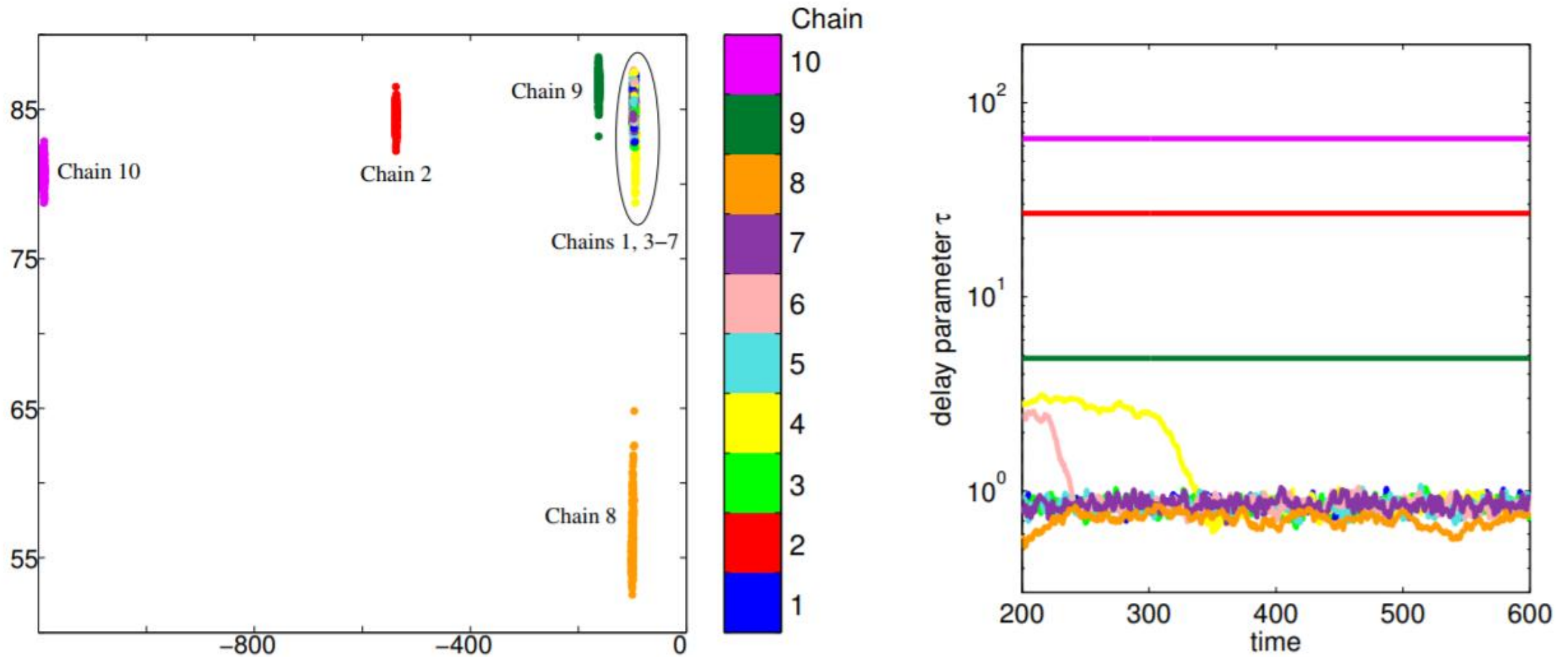
- Will a Gibbs sampler with Markov process defined on the right converge?
 - No, bipartite graphs are cyclic with even period
 - Consider transition matrix $T = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$
- And starting points $\left(\frac{1}{2}, \frac{1}{2}\right), (1,0)$



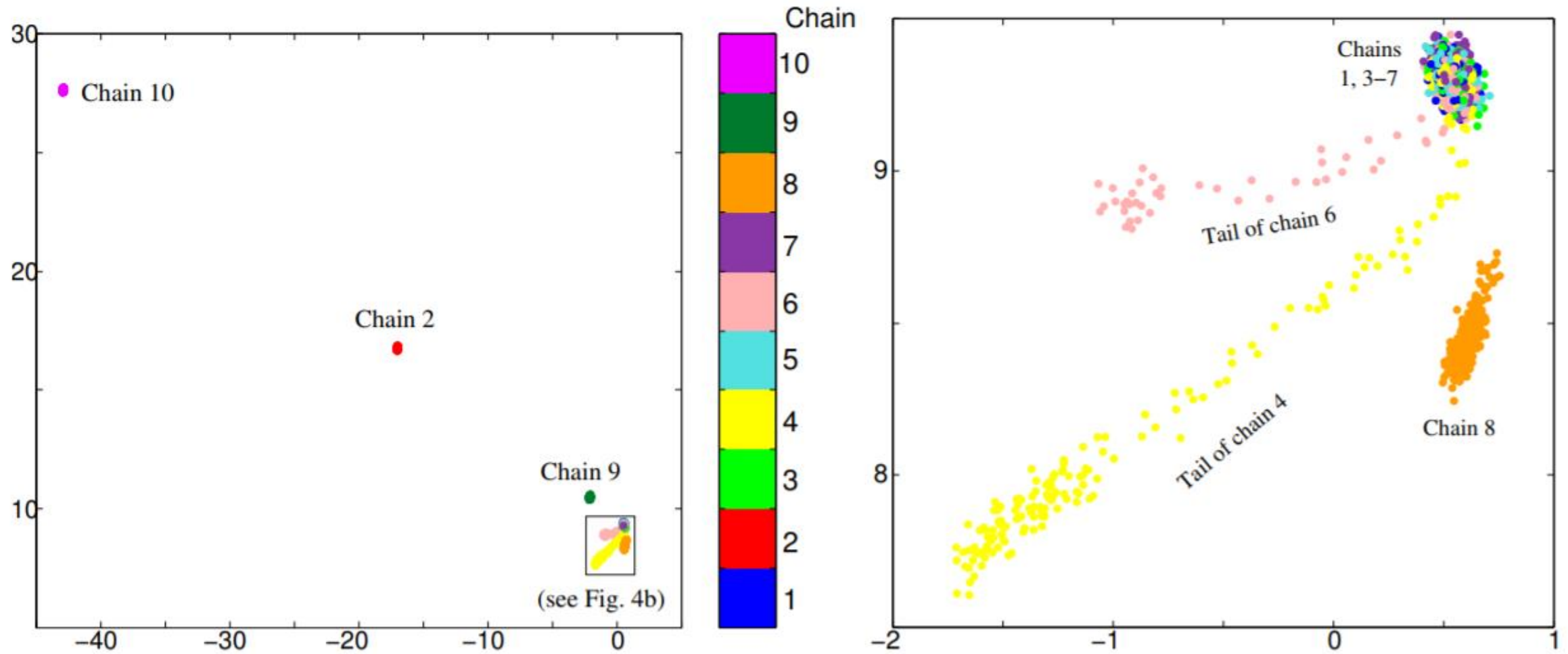
Why does this work? (in notes)

- **Ergodic or regular:** a MC is ergodic if it is **irreducible and aperiodic**
- **Theorem:** A regular Markov chain converges to a unique stationary distribution
- **Reversible:** chain transitions between x and x' are symmetric given $\pi(x)$
 - A reversible MC has stationary distribution $\pi(x)$
- **Theorem:** If a Markov chain is regular and satisfies reversibility with respect to $p(x)$, then $p(x)$ is its unique stationary distribution

Mixing times: linear discriminant analysis

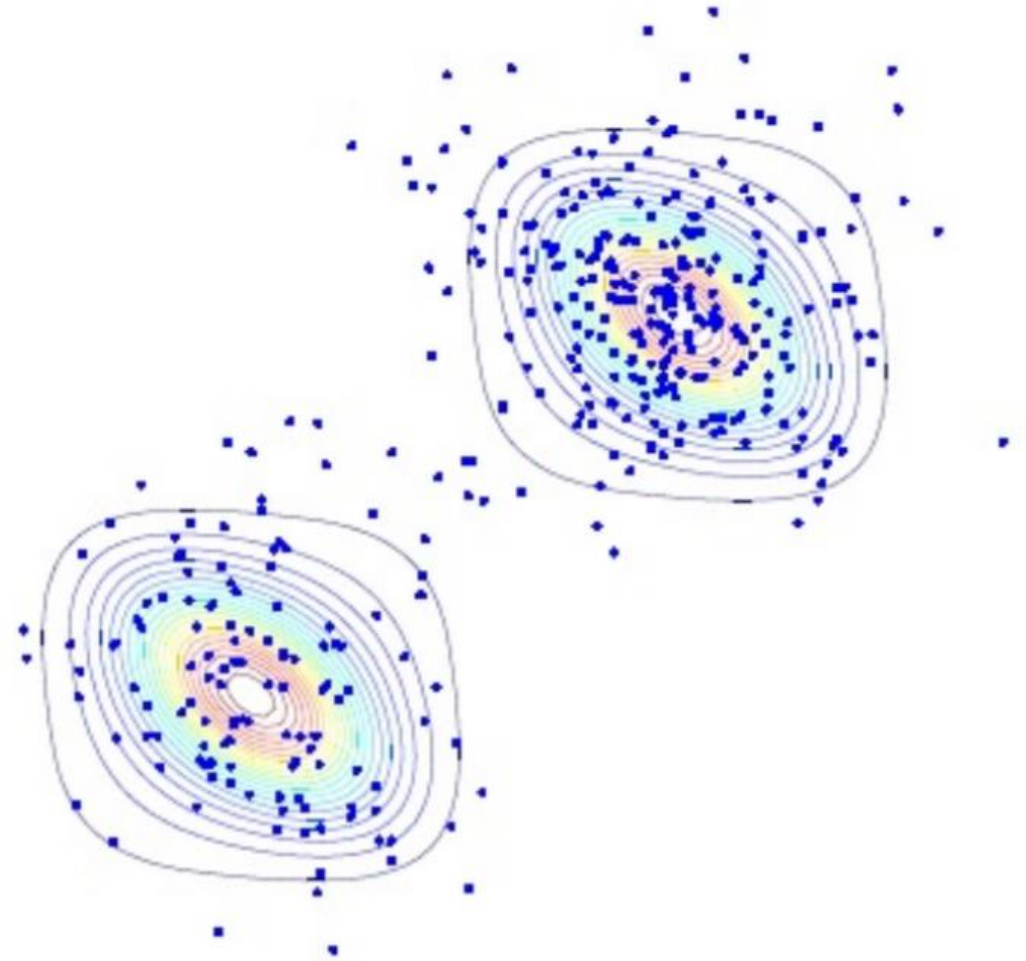


Mixing time: discriminative component analysis



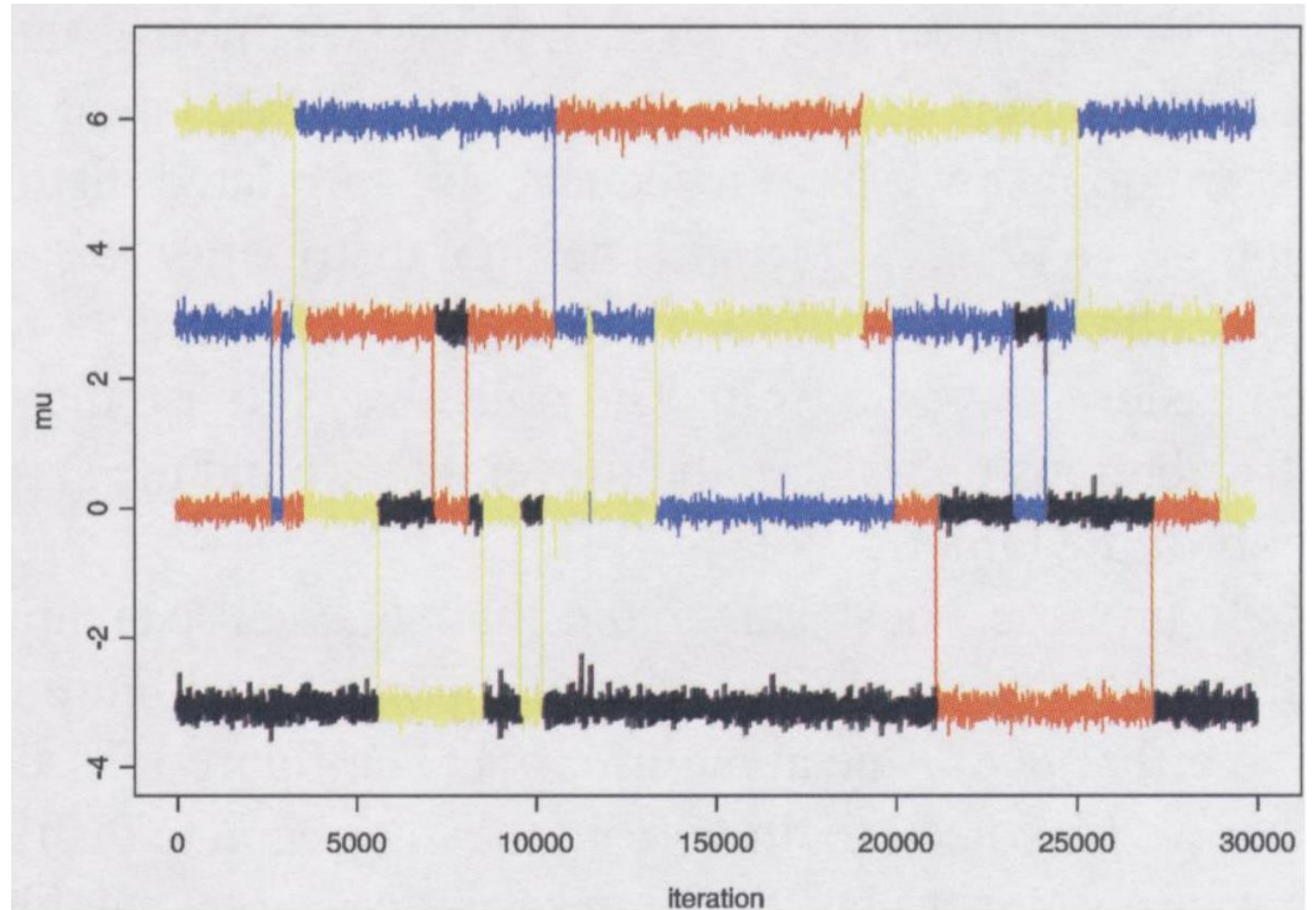
Problems with Gibbs sampling and MH

- Simulations from a multimodal distribution
- Samples from an MH run shown to the right
- Can you explain what is happening?



An issue with MCMC

- 4 component Gaussian mixture model
- Colors denote mean components ($\mu_1, \mu_2, \mu_3, \mu_4$)
- What is going on here?

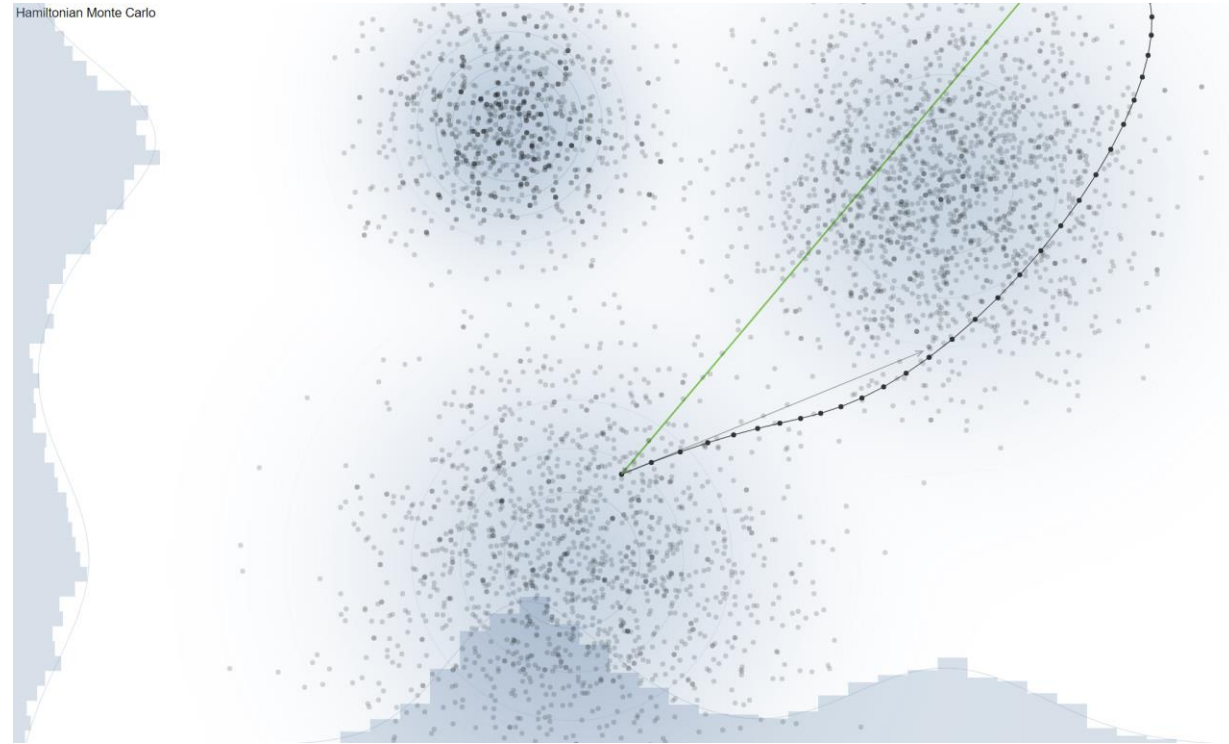


Variants of MCMC

- Slice Sampling
- Gibbs Sampling
- MH Sampling
- Adaptive MH Sampling
- No-U-Turn Sampling
- Metropolis-adjusted Langevin Algorithm
- Stein Variational Gradient Descent
- **Hamiltonian Monte Carlo**

In notes: Hamiltonian Monte Carlo

- Augment space with momentum variables
- Model variables modelled as position, momentum variables sampled from prior
- Choose a direction, use momentum to sample the next point



Markov Chain Monte Carlo summary

- General purpose method for Bayesian inference
- Often easy to implement
- Good theoretical guarantees
- Difficult to know when the chain has sufficiently mixed
- Can be very slow to converge
- Many tunable parameters

MCMC Summary

Wish to compute

$$p(\mathbf{z} | \mathbf{x}) = \frac{p(\mathbf{z}, \mathbf{x})}{p(\mathbf{x})}$$

Marginal is hard

$$p(\mathbf{x}) = \int p(\mathbf{z}, \mathbf{x}) d\mathbf{z}$$

1. Construct an ergodic and reversible MC on $\mathbf{z}$ whose stationary distribution is the posterior $p(\mathbf{z} | \mathbf{x})$
2. After mixing, collect samples from the stationary distribution
3. Approximate the posterior with an empirical estimate constructed from the collected samples.

